Operational Performance in Nordic Private Equity Buyouts

Operating Effects of Nordic Private Equity Buyouts: Matched Evidence on Growth, Profitability and Macroeconomic Heterogeneity

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Abstract

This thesis examines whether private equity (PE) buyouts are associated with operational improvements in Nordic portfolio companies and which factors explain differences in post-buyout performance. Using 226 completed buyouts in Sweden, Norway, Denmark and Finland between 1999 and 2023 (Bureau van Dijk's Orbis database), it combines a before-after event study with OLS regressions and a propensity score matching and difference-in-differences (PSM-DiD) design.

The central finding is a growth-over-margin pattern: buyouts are followed by significant revenue growth (+45 percent, $p < 0.001$) alongside a significant decline in EBITDA margins (-3.08 pp, $p < 0.01$), consistent with sponsors prioritising investment-phase capacity building over short-run efficiency. The PSM-DiD estimates attribute this pattern to PE ownership rather than to firm characteristics or economy-wide trends. The margin effect (-2.6 pp, $p < 0.05$) is robust to unobserved confounding, while the revenue effect (+14.3 pp, $p < 0.05$) is more sensitive and is best read as an upper bound.

Three moderators stand out: buyouts completed in the high-rate period from 2022 show the largest negative revenue effect (approximately 35 pp), though its magnitude is sensitive to the comparison period. Institutional buyouts outperform management buyouts on revenue growth (+39 pp, $p < 0.05$), and larger targets achieve greater margin improvements (+1.69 pp per log unit of revenue, $p < 0.01$). A supplementary buy-and-build sub-analysis indicates that platforms pursuing add-on acquisitions achieve this revenue growth without the margin compression seen in standalone investments, pointing to two distinct value-creation paths within the PE-backed sample.

The thesis contributes the first explicit test of the post-2021 interest-rate environment as a moderator of Nordic PE operational performance, within a causal PSM-DiD design.

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1. Introduction

The private equity (PE) industry has undergone a fundamental transformation over the past four decades. The leveraged buyout (LBO) model that emerged in the United States during the 1980s was initially characterised by financial engineering, specifically the disciplinary use of debt, ownership concentration, and incentive realignment, as the primary mechanism of value creation (Jensen, 1986; Kaplan, 1989). As credit markets developed, competition for deal flow increased and the supply of undervalued public targets declined. The industry gradually placed more emphasis on what Kaplan and Strömberg (2009) describe as "operational engineering": the direct improvement of portfolio company operations through managerial expertise, strategic repositioning, and enhanced governance. Understanding whether and how this operational value creation translates into measurable performance outcomes has become one of the central questions in corporate finance.

The Nordic private equity market provides an interesting setting to study this question. The region grew significantly over the previous decade, reaching a peak of 1038 transactions worth 93 billion EUR in 2021 before activity declined as interest rates increased the years thereafter. By 2024, deal activity had fallen by approximately 40 percent from its 2021 peak (Gain.pro, 2025; PitchBook, 2025). PE-backed businesses in Sweden, Norway, Denmark, and Finland employ together approximately one million people, representing roughly six percent of the working-age population, and PE penetration in the Nordics, at approximately 15 percent, exceeds the level in the rest of Europe of around 11 percent (Gain.pro, 2025). Nordic buyouts have historically outperformed buyouts in the rest of Europe and North America, with a favourable loss ratio (StepStone Group, 2023). According to StepStone Group's analysis of 573 realised Nordic transactions, value creation has primarily come from revenue growth (55 percent), followed by market multiple expansion (28 percent) and EBITDA margin expansion (18 percent) (StepStone Group, 2023). Taken together, these patterns indicate that operational improvement, rather than financial engineering alone, plays a central role in value creation in the Nordic region.

Yet, the operating conditions that defined Nordic PE performance through most of the 2010s changed abruptly from 2022 onward. After the post-pandemic rise in inflation, Riksbanken, Norges Bank, and the European Central Bank began tightening monetary policy aggressively, raising policy rates at a pace not seen in decades. For PE-backed companies, whose capital structures are typically characterised by significant leverage, this shift increases debt servicing costs, constrains the cash available for operational investment, reduces consumer demand, and compresses exit valuations. The KPMG Nordic PE market update (2025) notes that Nordic buyout activity came under pressure in 2023 following the rate cycle. Whether this macroeconomic regime shift has affected the operational

performance outcomes of PE ownership is a question that has received limited academic attention, as most existing studies of PE operational performance were conducted during periods of historically low and supportive interest rate conditions. This motivates the following research question:

Do private equity buyouts affect operational performance in Nordic portfolio companies, and what factors explain variation in post-buyout performance?

To answer this question, this thesis examines whether private equity buyouts are associated with measurable changes in operational performance of Nordic portfolio companies, and which deal-level and macroeconomic factors explain variation in post-buyout performance. Using a sample of 226 completed Nordic buyouts across Sweden, Norway, Denmark, and Finland from 1999 to 2023, sourced from Bureau van Dijk's Orbis database, we use a before-after event study design with a symmetric ±2-year window around the buyout date. Operational performance is measured through changes in EBITDA margin and revenue growth. To reduce the risk that observed changes are driven by broader economic trends or pre-existing firm characteristics as opposed to PE ownership itself, we draw matched controls from a pool of 4707 comparable non-PE-backed Nordic companies through propensity score matching and estimate the treatment effect using a difference-in-differences framework.

The analysis shows five main findings, with an additional buy-and-build sub-analysis. First, the univariate before-after analysis shows that PE buyouts are followed by a statistically significant decline in EBITDA margins of 3.08 percentage points alongside significant revenue growth of 45 percent in the post-buyout period, consistent with PE sponsors prioritising top-line growth over short-run profitability improvement.

Second, institutional buyouts (IBOs) are associated with significantly higher revenue growth than management buyouts (MBOs), at approximately 39 percentage points.

Third, buyouts completed during the high-interest-rate period from 2022 onwards are linked with approximately 35 percentage points lower post-buyout revenue growth relative to pre-2022 cohorts. This is the largest effect estimated in the study, though its statistical significance is sensitive to the choice of comparison period.

Fourth, larger firms at the time of acquisition show significantly greater EBITDA margin improvement post-buyout, in accordance with PE operational engineering producing scale-conditional returns.

Fifth, the propensity score matching and difference-in-differences analysis indicates that this growth-over-margin pattern, where revenue outperformance occurs alongside margin compression, is attributable to PE ownership rather than firm characteristics alone.

The supplementary buy-and-build sub-analysis shows that B&B (Buy-and-Build) platforms, defined as PE-backed companies that completed at least one add-on acquisition during the observation window, have significantly higher revenue growth relative to matched controls, but no significant margin compression. Standalone PE investments show the opposite pattern, significant margin compression, but no significant revenue outperformance. This difference points to two empirically distinct value-creation paths within the PE-backed sample, and suggests that the margin compression observed in the full sample is mainly driven by standalone investments rather than B&B platforms.

This thesis contributes to the literature in three ways. First, it presents empirical evidence on Nordic PE operational performance using a sample that extends to 2023, thereby broadening the time period and geographic coverage of existing Nordic-specific studies. Second, it is, to our knowledge, the first study to explicitly test the role of the post-2021 interest rate environment as a moderator of PE operational performance in the Nordic context, adding a new macroeconomic perspective to the literature. Finally, it employs a causal identification strategy, propensity score matching combined with difference-in-differences estimation, that addresses the selection bias concern central to the PE performance literature.

The rest of the thesis is organized as follows: Section 2 reviews the theoretical and empirical literature on PE value creation, operational performance, and the Nordic market. Section 3 outlines the data sources, sample construction, variable definitions, and empirical methodology. Section 4 presents the main empirical results, covering univariate statistics, OLS cross-sectional regressions, and the PSM-DiD analysis. Section 5 reports robustness checks. Section 6 discusses the findings in relation to the existing literature. Section 7 acknowledges the limitations of the study. Section 8 concludes.

2. Literature Review and Hypothesis Development

This chapter reviews the theoretical and empirical foundation of private equity operational performance research and derives five testable hypotheses for the Nordic context. The review proceeds from the agency-theoretic origins of the leveraged buyout model through empirical evidence on post-acquisition operating outcomes, before examining three sources of heterogeneity that motivate the predictions tested in this study: macroeconomic regime, deal type, and pre-buyout firm size. The chapter concludes by addressing the selection bias problem and specifying the causal

hypothesis that the propensity score matching and difference-in-difference framework is designed to test.

2.1 The Agency-Theoretic Foundation of the Buyout Model

The intellectual origins of the private equity model rest on Jensen's (1986) agency cost framework. Jensen (1986) observed that managers of cash-rich public corporations face systematic incentives to pursue empire-building and invest in negative net present value projects instead of returning capital to shareholders, reflecting the free cash flow problem created by the separation of ownership and control. The leveraged buyout addresses this conflict through three simultaneous mechanisms. First, the disciplinary effect of debt, which forces management to generate sufficient cash flows to service obligations and thereby curtails wasteful discretionary spending. Second, concentrated ownership, which eliminates the dispersed and largely passive shareholder structure that enables managerial rent-seeking. Third, high-powered managerial incentives, typically through significant equity co-investment, which directly align management interests with those of PE sponsors (Jensen, 1989).

Kaplan (1989) was among the first to test these predictions empirically, finding significant improvements in operating cash flow and reductions in capital expenditure for a sample of 76 management buyouts of U.S. public companies completed between 1980 and 1986. Later studies, including Lichtenberg and Siegel (1990), who found productivity improvements at PE-backed manufacturing plants, and Smith (1990) who documented operating efficiency gains across a broader MBO sample, provided further support for the governance hypothesis. The consensus from this early literature was that PE ownership constituted a superior organisational form for a certain class of mature, cash-generative businesses with limited remaining internal growth opportunities.

However, some critics argued that this early evidence reflected a historically unique context: a wave of hostile public-to-private transactions targeting chronically inefficient, over-diversified U.S. corporations, many of which had considerable room for restructuring. As the supply of such targets declined and competition for deal flow intensified, the way PE creates value also changed. Kaplan and Strömberg (2009) characterize this development as a shift from financial and governance engineering, based on leverage and ownership restructuring, toward what they describe as operational engineering: improving portfolio company operations through managerial expertise, strategic repositioning and operational improvements. This shift is also reflected in the aggregate value creation data, where leverage has contributed a declining share of buyout value creation over

time, while operational improvement has accounted for a growing share (Kaplan and Strömberg, 2009).

A broad overview of this development is provided by Verbouw, Meuleman, and Manigart (2025), whose meta-analysis of 66 empirical studies spanning over four decades documents a clear shift in PE value creation. In the twentieth century, buyouts were mainly related to significant positive effects on portfolio company efficiency, but not on growth. In the twenty-first century, this pattern has reversed. Efficiency effects have gradually weakened, while growth improvements have become the dominant outcome. Verbouw et al. (2025) attribute this shift to competitive pressures that made the financial engineering model less distinctive and forced PE firms to develop operational and strategic capabilities as alternative sources of value creation. This temporal pattern provides important context for the present study, which examines Nordic buyouts from 1999 to 2023 and therefore spans precisely the period in which this transition from efficiency to growth as the primary PE value creation mechanism has been most pronounced. This study focuses on the operational, growth-oriented channel of PE value creation. The disciplining role of leverage emphasised by Jensen (1986) cannot be tested directly here, because entity-level reported leverage does not capture acquisition-level (SPV/Bidco) debt structures, later described in Section 7.4.

2.2 Empirical Evidence on Post-Buyout Operational Performance

The broader empirical literature on PE operational performance is characterised by substantial heterogeneity in findings, reflecting differences in geography, sample period, deal type, and methodology. For the United States, Guo, Hotchkiss, and Song (2011) find that operating performance gains in public-to-private buyouts from the 1990s and 2000s are limited and largely attributable to industry trends as opposed to PE-specific interventions. Cohn, Mills, and Towery (2014), using confidential tax return data to study the near-universe of U.S. public-to-private buyouts between 1995 and 2007, report little evidence of systematic improvement in operating profitability on average. These findings suggest that the gains documented by Kaplan (1989) for 1980s transactions did not continue into the subsequent two decades, likely because the governance problems that PE targeted in that era had been greatly improved by improved public market governance.

In contrast, a growing literature on private-to-private transactions, the dominant deal type in European PE markets, reports more consistently positive operational outcomes. Boucly, Sraer, and Thesmar (2011) study 839 French buyouts and document substantial growth in revenues, employment, assets, and investment relative to matched control firms in the three years following a

buyout, attributing these effects to PE's role in relaxing financial constraints for credit-dependent firms that would otherwise be unable to finance profitable growth projects. Acharya, Gottschalg, Hahn, and Kehoe (2013) examine 395 European leveraged buyouts and find that operational performance is concentrated in transactions where the responsible PE partner has an operational background, indicating that this operational expertise is an important source of value creation. Bernstein and Sheen (2016) provide causal evidence from the U.S. restaurant industry, showing that PE-owned stores have stronger operational practices relative to franchised units of the same chain. Their identification approach reduces the selection bias concern by comparing units within the same parent organization.

Verbouw, Meuleman, and Manigart (2025) also find that institutional context plays an important role in post-buyout efficiency and growth outcomes. Countries with stronger investor protection and more developed capital markets are associated with larger PE-induced improvements. This makes the Nordic countries an interesting setting for the present study, as they are consistently ranked among the most transparent, well-governed, and investor-friendly markets in the world (Transparency International, 2023). The Nordic context therefore allows us to examine whether PE adds operational value beyond what the institutional environment already provides.

For the Nordic region specifically, Grubb and Jonsson (2007) provide early evidence from Sweden, finding generally positive operational outcomes using industry-adjusted performance measures based on Barber and Lyon (1996). Bloom and Van Reenen (2007) show that PE-backed companies are often among the best-managed in cross-country surveys of management practices, although they caution that this pattern may reflect pre-buyout selection, and not a causal effect of PE ownership. These findings, combined with StepStone Group's (2023) analysis of value creation in Nordic transactions, suggest that operational improvement is an important source of value creation in the Nordic region. This supports the view that the operational engineering thesis may apply more clearly in private-to-private transactions than in the large public-to-private deals that dominate much of the U.S. literature.

Against this background, the first hypothesis concerns the average operational trajectory of Nordic PE-backed companies following a buyout, viewed as a before-after comparison.

H1: Overall Operational Performance

Nordic PE-backed companies experience a statistically significant decline in EBITDA margin in the post-buyout period (t_{+1}, t_{+2}) relative to the pre-buyout period (t_{-2}, t_{-1}), consistent with investment-phase margin compression under a growth-oriented ownership model.

The expected direction of H1 is negative, predicting a short-run decline in EBITDA margins after the buyout. This prediction follows from the growth LBO hypothesis articulated by Boucly, Sraer, and Thesmar (2011), who show that PE ownership relaxes financial constraints for credit-dependent private firms and allows them to invest in growth projects that may reduce short-run margins while increasing revenue growth. In the Nordic private-to-private context, where PE targets are mainly small and medium-sized enterprises, and not just mature public corporations, value creation is expected to come primarily through capacity building as opposed to short-term efficiency gains. This capacity building includes additional headcount, market entry expenditure, and management professionalisation. These investment-phase costs are expected to compress EBITDA margins within the ±2-year observation window, before the related revenue gains are fully realised. The negative prediction is also supported by the StepStone Group's (2023) decomposition of Nordic PE value creation, in which revenue growth dominates margin expansion. Jensen's (1986) efficiency-extraction prediction of positive margin change is therefore considered the theoretically weaker prior in the Nordic private-to-private context, and H1 is stated with a negative expected direction accordingly.

2.3 The Interest Rate Channel: Macroeconomic Conditions as a Moderator

A dimension of the PE performance literature that has received limited systematic attention is the role of the prevailing interest rate environment in shaping post-buyout operational outcomes. Axelson, Jenkinson, Strömberg, and Weisbach (2013) provide important empirical evidence on this issue, showing that buyout leverage is driven more by economy-wide credit market conditions than by company-specific fundamentals. Specifically, leverage multiples in PE transactions move with the cost of credit across the business cycle, increasing when debt is easily available and inexpensive, and contracting when the credit conditions tighten. This pattern implies systematic variation in the financial risk embedded in deal cohorts across time. The implication of this finding is that deal cohorts completed during periods of cheap credit inherit capital structures that become much more burdensome when interest rates later rise.

The post-pandemic monetary tightening cycle that began in 2022 represents arguably the most abrupt and severe macroeconomic regime change affecting PE-backed companies in decades. Riksbanken, Norges Bank, and the European Central Bank all raised policy rates at a pace not observed since the early 1980s, increasing the cost of the leveraged capital structures that support buyout returns. For PE-backed portfolio companies, this shift affects operational investment, research and development and expansion initiatives after interest obligations are met. More broadly, rising rates can reduce consumer demand and tighten credit conditions across supply chains and customer

bases, thereby constraining revenue growth. The KPMG Nordic PE market update (2025) notes that buyout activity came under pressure in 2023 following the rate cycle, with deal volumes declining approximately 40 percent from their 2021 peak (Gain.pro, 2025).

Despite the theoretical importance of this channel, and despite the magnitude of the 2022 - 2023 monetary tightening, existing academic studies of PE operational performance have been mainly conducted during the period of low and supportive monetary policy from 2010 to 2021. As a result, they offer limited insight into PE performance in a genuinely high-rate environment. This leaves an important gap in the literature, which the present study addresses by defining a separate high-rate period indicator for deals completed in 2022 or later. The second hypothesis follows directly from the theoretical channels discussed above:

H2: Interest Rate Environment

Buyouts completed in 2022 or later are associated with significantly lower post-buyout revenue growth compared to buyouts completed before 2022, reflecting the operational and financial constraints imposed by the post-pandemic monetary tightening cycle.

The expected direction of H2 is negative: deals completed during the high-rate period are expected to face less favorable conditions for revenue growth, both through the direct cost-of-capital channel identified by Axelson et al. (2013) and through the wider macroeconomic mechanisms described above. The hypothesis is tested using a high-rate dummy variable in the cross-sectional OLS regressions. The coefficient captures the difference in revenue growth between post-2021 deals and earlier cohorts, after controlling for firm size, initial margin, deal type, and country fixed effects.

2.4 Deal Type Heterogeneity: Institutional versus Management Buyouts

A second source of heterogeneity in post-buyout operational outcome concerns the nature of the transaction itself, specifically whether the buyout is led by a financial institution (an IBO, in which the acquiring entity is a PE fund or financial sponsor) or by the incumbent management team (an MBO). This distinction is important because the two deal types differ notably in the financial resources, operational expertise, and strategic ambition that the new owners bring to the portfolio company.

In an institutional buyout, a PE sponsor acquires a controlling stake, often as a secondary buyout from another PE fund, and implements a structured value creation plan that is typically developed during due diligence. The acquiring sponsor brings not only capital but also operational resources developed through prior investments, including proprietary management frameworks, networks of operational advisors, and experience with sector-specific value creation levers. Secondary buyouts are particularly

relevant in the Nordic market, as they represent a large share of IBOs. Acharya et al. (2013) document that general partners with operational experience (ex-consultants or ex-managers) generate stronger operational improvements. This is relevant because secondary buyout sponsors often target assets where they have accumulated domain expertise through previous platform investments.

Management buyouts, by contrast, are led by the existing management team, typically in partnership with external financing and modest PE investment. While MBOs align managerial incentives directly with ownership, a prediction of Jensen's (1986) governance framework, the incumbent management team may lack access to the same depth of operational resources and external expertise that institutional sponsors bring. The MBO structure may also preserve strategic continuity, but this can limit the scope for major operational change. This difference in available resources suggests that IBOs are more likely to pursue ambitious growth strategies and achieve stronger revenue expansion than MBOs. In MBOs, the primary mechanism of value creation is more likely to be incentive realignment rather than strategic repositioning.

This reasoning motivates the third hypothesis, which concerns the differential revenue growth outcomes between IBOs and MBOs in the Nordic sample.

H3: Institutional versus Management Buyouts

Institutional buyouts (IBOs) generate significantly higher post-buyout revenue growth than management buyouts (MBOs), reflecting the greater operational resources, strategic ambition, and growth-oriented ownership model associated with PE sponsor involvement.

The expected sign of the IBO coefficient in the OLS regression is positive, in line with the hypothesis that financial sponsors use more active growth-oriented ownership approaches than management-led acquisitions. This hypothesis is tested using a binary IBO indicator in the cross-sectional regression framework. The coefficient captures the difference in revenue growth associated with deal type after controlling for size, profitability, macroeconomic period, and country effects.

2.5 Pre-Buyout Firm Size and Scale-Conditional Returns to Operational Engineering

A third potential source of heterogeneity in post-buyout outcome concerns the size of the portfolio company at the time of the acquisition. The relationship between firm size and PE induced operational improvement is theoretically ambiguous and empirically contested, but two distinct mechanisms generate opposite predictions that warrant careful examination.

The first mechanism relates to managerial capacity constraints. Smaller, often privately held or family-owned companies frequently exhibit less formalised management practices, including diffuse decision-making, limited use of formal performance metrics, and underdeveloped governance structures. This is often because their owners lack the expertise, incentives, or resources to implement professional management frameworks. Bloom and Van Reenen (2007) show substantial variation in management quality across firm size in cross-country surveys, with smaller firms more likely to score lower on structured management practices. PE ownership can address this deficit by professionalising governance and introducing best-practice management frameworks, which may generate particularly large improvements in smaller targets where the initial management gap is greatest.

The second mechanism operates through economies of scale in operational engineering. Larger firms, given their size, may offer more scope for process optimisation, cost rationalisation, and cross-functional restructuring, and PE sponsors may find it easier to implement standardised value creation playbooks in targets with more developed organisational structures. This would predict the opposite relationship: operational improvements concentrated in larger targets. Which of these two mechanisms dominates is ultimately an empirical question, as the existing evidence is mixed and offers no clear common view on whether operational gains scale with target size.

In the Nordic context, the predominance of small and medium-sized enterprises and the private-to-private character of the buyout market, which is consistent with the evidence in Boucly et al. (2011), suggests that EBITDA margin improvement may be particularly sensitive to pre-buyout firm size, with larger portfolio companies offering more scope for operational engineering initiatives. The fourth hypothesis formalises this prediction, tested through the size coefficient in the OLS regression.

H4: Pre-Buyout Firm Size and EBITDA Margin Improvement

Larger firms at the time of buyout, as measured by pre-buyout operating revenue, experience significantly greater post-buyout EBITDA margin improvements, consistent with PE operational engineering producing scale-conditional returns in the Nordic private-to-private context.

The expected sign of the size coefficient is positive, reflecting the hypothesis that margin improvement is increasing in firm size. This prediction is aligned with a view in which larger targets offer more scope for identifiable cost reduction, process optimisation, and professionalisation initiatives that translate into margin expansion, while very small targets see compression as PE sponsors invest in building organisational capacity ahead of revenue realisation.

2.6 Selection Bias and the Causal Effect of PE Ownership

A fundamental methodological challenge in the PE performance literature is that observed post-buyout improvements may reflect the selection of high-quality targets instead of a causal effect of PE ownership. PE sponsors invest substantial resources in identifying, screening, and performing due diligence on potential portfolio companies, and PE funds are known to target high-potential firms with strong fundamentals and clear improvement opportunities (Jenkinson et al., 2022). Kaplan and Strömberg (2009) describe how PE firms use their industry and operating knowledge to identify attractive investments. If these pre-existing characteristics also explain stronger post-buyout performance, a simple before-after comparison may mix selection effects with treatment effects and overstate PE's operational contribution. Jenkinson et al. (2022) identify this selection concern as a central methodological challenge in the field, because sponsors deliberately acquire high-potential targets, it is difficult to separate the operational value created by PE owners from the performance these firms might have achieved under another ownership structure.

The propensity score matching approach, introduced to PE literature by Boucly et al. (2011) and applied in a Nordic context in this study, addresses this concern by constructing a matched control group of comparable non-PE-backed companies that resemble the treated firms based on observable pre-buyout characteristics. By conditioning on observable selection factors, including pre-buyout firm size, initial EBITDA margin, and country of origin, the matched comparison comes closer as an as-if randomised experiment conditional on observables. The subsequent difference-in-differences estimation compares changes in outcomes over time between treated firms and matched controls. This helps account for time-invariant unobserved heterogeneity and isolates the part of observed performance change that can be attributed to PE ownership itself. Together, this two-stage identification strategy provides the most credible evidence on whether Nordic PE-backed companies outperform what would have been expected without the buyout.

The methodological importance of controlling for selection effects is further supported by Verbouw et al. (2025), whose meta-regression results indicate that studies using matching methods tend to report smaller effect sizes than studies comparing PE-backed firms to unmatched control groups. This pattern is likely driven by non-random target selection in PE buyouts. This finding supports the present study's use of propensity score matching as the main identification strategy and suggests that the ATT estimates reported in Section 4 should be interpreted as more conservative and credible than estimates from unmatched before-after designs.

The fifth and final hypothesis formalizes the prediction about the causal PE effect, as distinct from the descriptive before-after comparisons captured in H1 through H4.

H5: Causal Effect of PE Ownership

Relative to propensity-score matched non-PE-backed peers, PE-backed Nordic companies show significant revenue outperformance alongside margin compression in the post-buyout period, demonstrating that these operational differences are attributable to PE ownership rather than to pre-existing firm characteristics.

2.7 The Buy-and-Build Strategy as a Distinct Value Creation Pathway

A particularly relevant PE operational strategy in the Nordics is the B&B (Buy-and-Build) approach, in which a PE-backed platform company makes a series of acquisitions of smaller targets to consolidate a fragmented market segment. Hammer, Knauer, Pflücke and Schwetzler (2017) document B&B as a distinct and widespread PE strategy, finding that add-on activity is more likely when the PE sponsor is experienced, the platform firm is large, and the target industry is moderately fragmented. On the performance side, Cohn, Hotchkiss, and Towery (2022) find that buyouts undertaking add-ons achieve stronger revenue growth, while Bansraj, Smit, and Volosovych (2020) show that B&B platforms achieve higher margins and labour profitability than standalone PE investments, which they attribute to operational synergies between the combined companies.

The Nordic PE market is characterised by high B&B activity. Gain.pro, (2025) documents that 54 percent of Nordic portfolio companies pursue buy-and-build strategies, well above the European average, reflecting the fragmented structure of many Nordic mid-market sectors. This widespread use has important implications for interpreting aggregate Nordic PE performance metrics, as a meaningful share of post-buyout revenue growth may come from the consolidation of acquired revenues compared to organic operational improvement.

The operational implications of B&B activity for margin and revenue outcomes are theoretically unclear. On the revenue side, inorganic growth is expected to strengthen top-line performance for B&B platforms relative to standalone investments. On the margin side, integration costs, management capacity absorbed by deal execution, and the typically lower initial margins of acquired targets may offset efficiency gains, leading to slower margin growth or margin compression during the active build phase. These predictions motivate the exploratory sub-analysis described in Section 3.12 and reported in Section 6.4.

Table 1: Summary of Hypotheses

Hypothesis	Statement	Empirical Test	Expected Sign
H1	Nordic PE-Backed companies show a statistically significant decline in EBITDA margin in the post-buyout period	Paired t-test and Wilcoxon signed-rank test on ΔEBITDA margin (pp)	**Negative (margin compression)**
H2	Buyouts completed in 2022 or later are associated with significantly lower post-buyout revenue growth compared to buyouts completed before 2022.	High_rate dummy in OLS regression (%Δ operating revenue)	**Negative**
H3	Institutional buyouts generate higher post-buyout revenue growth than management buyouts.	IBO dummy in OLS regression (%Δ operating revenue)	**Positive**
H4	Larger firms at the time of buyout experience greater post-buyout EBITDA margin improvement.	log_size coefficient in OLS regression (ΔEBITDA margin)	**Positive**
H5	Relative to matched non-PE-backed peers, PE-backed Nordic firms show significant revenue outperformance alongside margin compression	PSM + DiD ATT estimate (ΔEBITDA margin, %Δrevenue)	**Mixed (+ Revenue, - Margin)**

Note: H1 is tested via paired t-test and Wilcoxon signed-rank test on the treatment sample (N = 226). H2-H4 are tested via OLS cross-sectional regressions with HC1 robust standard errors. H5 is tested via PSM-DiD estimation on the matched sample (226 PE-backed firms matched 3:1 to non-PE-backed controls, N = 904 observations), with standard errors clustered at the firm level. All hypotheses use a symmetric ±2-year event window around the buyout date.

3. Data and Methodology

3.1 Research Design

This study uses a multi-method empirical approach to examine whether private equity (PE) ownership is associated with operational improvements in Nordic portfolio companies. The empirical strategy is structured in three stages. First, a before-after univariate analysis documents changes in operational performance for the treatment group across a ±2-year window around the buyout date. Second, OLS cross-sectional regressions identify deal-level and macroeconomic factors that explain differences in post-buyout performance. Third, a propensity score matching (PSM) procedure combined with a difference-in-differences (DiD) framework is used to estimate the causal effect of PE ownership by comparing treated firms with a matched control group of comparable non-PE-backed Nordic companies. Together, these three approaches allow us to describe the average performance path of PE-backed firms, explain variation in outcomes, and assess whether observed changes are attributable to PE ownership instead of pre-existing firm characteristics or economy-wide trends.

The choice of a symmetric ±2-year event window reflects both theoretical and empirical considerations. While a longer post-buyout window of three or four years would capture a greater share of the operational improvements documented in the literature, an asymmetric or extended window creates a fundamental tension with the identification of macroeconomic period effects. Specifically, a t-1:t+3 window requires post-buyout data through 2025 for deals completed in 2022, which falls outside Orbis coverage available at the time of writing. Under this specification, fewer than two deals completed in 2022 or later have sufficient data coverage to be included in the high-rate cohort, making the high-rate dummy effectively collinear and the test of H2 statistically infeasible. The ±2-year window retains 32 high-rate era observations, providing enough observations to estimate the macroeconomic moderation effect. However, the limited subsample size means the causal estimates for this cohort should be interpreted as directional rather than conclusive. The trade-off between window length and high-rate cohort coverage is therefore resolved in favour of the specification that preserves the study's most distinctive and policy-relevant hypothesis test.

3.2 Sample Construction and Data Sources

The primary deal-level data source is Orbis M&A, which is used to identify completed Nordic PE buyout transactions. These deal observations are then linked to company-level financial data in Bureau van Dijk's database using each company's BvD ID. Orbis provides financial statement data and ownership information for private companies across Europe. Nordic PE buyouts are identified by

filtering for completed ownership transactions in Sweden, Norway, Denmark, and Finland where the deal type is classified as an institutional buyout (IBO) or management buyout (MBO). The initial sample encompasses 271 PE buyout events completed between 1999 and 2023. Observations are excluded if financial data for the full event window (±2 years) is unavailable or if the company completed a secondary transaction within the observation period. This leaves a final treatment sample of 226 buyouts.

For each portfolio company, annual data on operating revenue, EBITDA, EBITDA margin, and number of employees are collected for the two fiscal years prior to the buyout (t-1, t-2) and the two fiscal years following the buyout (t+1, t+2). The buyout year itself (t=0) is excluded from the performance windows to avoid capturing mechanical effects of the transaction on reported financials. All financial variables are measured in nominal local currency terms.

3.3 Add-on and Buy-and-Build Data

To identify platform companies pursuing a buy-and-build (B&B) strategy, a supplementary dataset was constructed from Orbis M&A using the platform deal structure filter set to “Buy & Build”, combined with a Nordic geography restriction (Sweden, Norway, Denmark, Finland). This search returned 1780 completed transactions, spanning from 1997 to 2024. The dataset includes the acquiror BvD ID number, which identifies the platform company executing each add-on acquisition.

The acquiror BvD ID from this dataset was then cross-matched against the BvD IDs of the 226 platform companies in the thesis sample. A platform company was classified as B&B acquiror if it appeared as the acquiror in at least one completed add-on transaction in the Orbis B&B dataset. This procedure identifies 77 of 226 platform companies (34.1%) as active B&B acquirors. The remaining 149 companies (65.9%) were classified as standalone PE investments with no verifiable add-on activity. This classification may understate the true B&B share, as some platform acquisitions may have been excluded through special purpose vehicle (SPV) structures with separate BvD IDs. The true B&B share is therefore likely higher than the 34.1% identified here.

3.4 Variables and Measures

Three operational performance metrics serve as outcome variables. The primary measure is the change in EBITDA margin, defined as the difference between the post-buyout and pre-buyout period average EBITDA margin, expressed in percentage points. The second measure is the percentage change in operating revenue between the pre- and post-buyout periods. The third is the percentage change in EBITDA. These measures follow the approach of Kaplan (1989) and Acharya et al. (2013),

who document post-buyout operating performance using accounting-based metrics that are available for privately held companies.

The OLS cross-sectional analysis uses five main explanatory variables in the baseline model, with pre-buyout leverage added in extended specifications. Pre-buyout firm size is measured as the natural logarithm of operating revenue at t=0. Initial EBITDA margin captures the pre-buyout profitability level and controls for mean reversion effects. Pre-buyout leverage (debt-to-assets ratio at t=0, winsorized at the 1st and 99th percentiles) is included in extended model specifications as a robustness check and not in the baseline model because of more limited data coverage. Two macroeconomic period dummies are included: a low-rate-era indicator equal to one for buyouts completed between 2017 and 2019, covering the period of historically low interest rates, and a high-rate-era indicator equal to one for buyouts completed in 2022 or later, capturing the post-pandemic monetary tightening cycle. Deal type is captured by a binary IBO indicator equal to one for institutional buyouts and zero for management buyouts. IBO classification is based on the deal type field in Orbis, where deals labeled “Institutional buy-out” in any variant are coded IBO=1 and deals labeled “Management buy-out” or “MBO” are coded IBO=0. Country fixed effects use Sweden as the reference category, and sector fixed effects use Other Services as the reference category.

For a subset of 219 buyouts with sufficient employee records in Orbis, revenue per employee is calculated for both the pre- and post-buyout periods as operating revenue divided by the number of employees. The percentage change in revenue per employee serves as an auxiliary measure of labor productivity, used in Section 6.1, helping differentiate between efficiency-gains and capacity-building explanations for the observed margin and revenue changes. This measure is not included in the primary OLS or PSM-DiD specifications due to the reduced and potentially non-random subsample coverage, and is reported descriptively only. Detailed variable definitions are provided in Appendix Table A.2. Pairwise correlations among the covariates are reported in Appendix Table A.5 and are low, indicating that multicollinearity is not a concern in the regression specification.

3.5 Univariate Analysis

The before-after analysis tests whether the mean and median changes in EBITDA margin, revenue growth, and EBITDA across the treatment sample is statistically different from zero. Statistical significance is tested using both a paired t-test and a Wilcoxon signed-rank test. The paired t-test assumes that the distribution of changes is approximately normal, while the Wilcoxon test is

non-parametric and less sensitive to outliers and skewed distributions. Reporting both tests helps ensure that the results are not driven by specific distributional assumptions.

3.6 Cross-Sectional OLS Regression

To identify the drivers of heterogeneity in post-buyout performance, we estimate the following cross-sectional OLS model:

$$\Delta Performance_i = \alpha + \beta_1 log\ size_i + \beta_2 initial\ margin_i + \beta_3 leverage_i + \beta_4 low\ rate_i + \beta_5 high\ rate_i + \beta_6 IBO_i + \gamma_c + \gamma_s + \varepsilon_i$$

where $\Delta Performance_i$ is the change in EBITDA margin, percentage change in operating revenue, or percentage change in EBITDA for company *i*. The model includes country fixed effects γ_c and sector fixed effects γ_s. The leverage variable (debt-to-assets ratio) is included only in extended specifications and is winsorized at the 1st and 99th percentiles to limit the influence of extreme values. Standard errors are heteroskedasticity-robust using the HC1 correction to account for variance heterogeneity across deals of different sizes and time periods.

3.7 Propensity Score Matching

To address the concern that PE funds selectively target companies that would have improved regardless of ownership, we construct a matched control group from a pool of 4707 comparable non-PE-backed Nordic companies identified in Orbis. These companies satisfy the same Nordic country filter (Sweden, Norway, Denmark, Finland), minimum revenue threshold, and independence criteria as the treatment sample, but are not identified as PE-backed over the observation period. To keep the event windows comparable over time, matching is performed separately within each deal-year cohort, so that every treated firm is matched only to control firms observed over the same calendar window. Because a control firm may satisfy the data requirements for more than one deal year, it can enter the matching pool in multiple cohorts. The propensity-score estimation sample therefore exceeds the number of unique control firms in the cleaned pool.

The propensity score is estimated via a logit model regressing a binary PE treatment indicator on pre-buyout firm size (log operating revenue), EBITDA margin, and country fixed effects. Each treated firm is then matched with three control firms using nearest-neighbour matching without replacement and a caliper of 0.20 standard deviations of the propensity score, resulting in a 3:1 matched sample. The matching procedure produces no unmatched treated observations. The post-matching standardised mean difference (SMD) is -0.120 for log firm size, just above the conventional 0.10 threshold recommended by Austin (2011), and -0.030 for initial EBITDA margin. The residual size

imbalance is directly controlled for in the DiD specification through the log operating revenue covariate, reducing its potential influence on the ATT estimate. The full propensity-score model estimates are reported in Appendix Table A.3.

3.8 Difference-in-Differences Estimation

The causal effect of PE ownership is estimated using the following difference-in-differences specification for the PSM-matched sample:

$$\Delta Y_i = \alpha + \beta PE_i + \theta' X_i + \gamma_c + \gamma_s + \lambda_y + \varepsilon_i$$

Where ΔY_i is the within-firm change in the operational performance metric from the pre- to post-buyout period (the change in EBITDA margin in percentage points, or the percentage change in operating revenue) for firm *i*, PE_i is an indicator equal to one for PE-backed firms and zero for their matched controls. X_i is a vector of pre-buyout covariates (initial EBITDA margin and log operating revenue), and γ_c, γ_s and λ_y denote country, sector, and deal-year fixed effects. Because the outcome is measured as the within-firm pre-to-post change, the coefficient β on the treatment indicator is itself the difference-in-differences estimate: it compares the average change of PE-backed firms with that of their matched controls and is interpreted as the average treatment effect on the treated (ATT). This first-differenced formulation is equivalent to a two-period panel difference-in-differences but removes all time-invariant firm heterogeneity by construction, so firm fixed effects are not separately estimated. Selection on observables is instead handled through the PSM matching weights. The model is estimated on the matched sample of 226 PE-backed firms and 678 matched controls (N = 904), with standard errors clustered at the firm level.

The causal interpretation of β relies on three identifying assumptions. The stable unit treatment value assumption (SUTVA) requires that the PE ownership of one firm does not affect the performance of other firms in the sample, a plausible assumption in the mid-market buyout context where portfolio companies mainly operate in distinct product markets with limited direct competitive overlap. The conditional independence assumption (CIA) requires that, conditional on the covariates used in the propensity score model (pre-buyout firm size, initial EBITDA margin, and country), treatment assignment is as good as random within the matched sample. This is the main motivation for the PSM procedure. By constructing a control group that is observationally similar to the treatment group at t=0, the matched DiD design reduces selection bias linked to observable firm characteristics. Unobservable factors, such as concealed growth or management quality, remain a residual concern

and are discussed in Section 7. Finally, the parallel trends assumption requires that treated and matched control firms would have moved along comparable outcome paths had the buyouts not occurred, supplying the counterfactual against which post-buyout divergence is measured. The empirical plausibility of this assumption is assessed in Section 5.3.

The two key outcome variables in the DiD analysis are Δ EBITDA margin (in percentage points) and percentage change in operating revenue. The percentage change in EBITDA is reported in the univariate and OLS analyses but is excluded from the DiD estimation because of its high variance and lack of statistical significance in the univariate tests (t-test p=0.234, Wilcoxon p=0.335), which would reduce the precision of the ATT estimate without adding analytical value.

3.9 Subgroup Analysis by Macroeconomic Period

The pre-2022 subsample includes buyouts completed before 2022 (n = 194), covering deals completed under periods of low or neutral interest rates. The high-rate subsample includes all buyouts completed in 2022 or later (n = 32), covering deals completed during the post-pandemic monetary tightening cycle.

Each subsample is re-matched independently using the same nearest-neighbor 3:1 procedure and caliper of 0.20 standard deviations described in Section 3.7, and the DiD specification from Section 3.8 is estimated separately within each matched subsample. This allows the ATT for both EBITDA margin and revenue growth to differ across macroeconomic periods and provides a direct causal test of whether the high-rate environment limits the operational value creation observed in the full-sample estimates. Given the limited size of the high-rate subsample, these subgroup estimates should be interpreted as directional and not definitive. With 32 matched treated observations, the statistical power to detect treatment effects of the size estimated in the full-sample analysis is low.

Using the full-sample revenue ATT of approximately 14 percentage points as a reference effect size, and scaling the standard error proportionally to the smaller sample, a post-hoc power calculation suggests the high-rate subsample has approximately 25 to 30 percent power to detect an effect of this size at the five percent significance level. This is well below the conventional 80 percent threshold. The 95 percent confidence interval for the high-rate revenue ATT, based on the point estimate of -17.7 percentage points and an implied standard error of approximately 16 percentage points, ranges from approximately -48 to +13 percentage points. This wide interval does not provide a precise estimate of the true effect size, but it is in accordance with a large negative effect, which the subsample is too small to estimate reliably. The lack of statistical significance at p = 0.257 should

therefore not be interpreted as evidence against the macroeconomic moderation hypothesis, but rather as a reflection of limited statistical power.

3.10 Parallel Trends Assumption

The validity of the DiD estimator relies on the parallel trends assumption. This requires the matched control firms to provide a credible approximation of how the PE-backed firms would have developed in absence of the buyout. In practice, the treated and control firms should show comparable performance trends before treatment. Since the post-buyout counterfactual cannot be observed, the assumption is evaluated indirectly using pre-treatment data. These checks are reported in Section 5.3.

3.11 Robustness Checks

The main findings are subjected to two robustness checks. First, the analysis is repeated excluding buyouts completed before 2010 (n=7), reducing the potential influence of the early-sample period, which may be characterised by different PE market dynamics and thinner control pools. Second, the PSM procedure is repeated using a stricter caliper of 0.15 standard deviations. This makes the matched pairs more comparable on observed pre-buyout characteristics, but may also leave some treated observations unmatched. Consistent results across both specifications increase confidence in the robustness of the main findings. In addition, the credibility of the identification strategy is examined through a parallel-trends assessment in Section 5.3, and a Rosenbaum sensitivity analysis of the estimates' robustness to unobserved confounding in Section 5.4.

3.12 Buy-and-Build Sub-Analysis

The buy-and-build (B&B) strategy, where a PE-backed platform company makes a series of acquisitions of smaller targets to consolidate a fragmented market, has been identified in the literature as a distinct value-creation pathway (Hammer, Knauer, Pflücke, & Schwetzler, 2017; Bansraj et al., 2020). To examine if B&B activity is associated with different operational outcomes within the PE-backed sample, three complementary analyses are conducted.

First, a descriptive comparison tests whether mean ΔEBITDA margin and percentage revenue growth differ between B&B platforms (≥1 confirmed add-on) and standalone PE investments, using a two-sample Welch t-test. Second, an OLS regression within the PE group estimates the conditional B&B effect:

$$\Delta Outcome_i = \alpha + \beta_1 \times BB_i + \beta_2 \times AddOnCount_i + \gamma \times X_i + \varepsilon_i$$

Where BB_i is a binary indicator for B&B platform status, $AddOnCount_i$ is the number of verified add-ons and is used as a continuous measure of acquisition intensity, and X_i is the standard control vector (margin_t0, log_size, year, country, and sector fixed effects). Heteroskedasticity-robust standard errors (HC1) are used for the within-PE OLS specification, while the DiD subgroup estimates use standard errors clustered at the firm level.

Finally, a DiD subgroup analysis estimates ATT separately for B&B platforms and non-B&B platforms relative to their matched control companies, using the same PSM-weighted DiD estimator described in sections 3.7 and 3.8. This makes it possible to assess whether PE vs. control treatment effect differs by acquisition strategy. Since the B&B classification is determined by post-investment behavior rather than random assignment, the within-PE OLS estimates should be interpreted as conditional associations based on pre-deal controls rather than causal effects.

4. Empirical Results

This section presents the empirical findings in three stages, following the research design outlined in Section 3. Section 4.1 reports descriptive statistics on the treatment sample and documents the average before-after changes in operational performance. Section 4.2 presents the OLS cross-sectional regression results, which identify the deal-level and macroeconomic determinants of heterogeneity in post-buyout outcomes. Section 4.3 reports the propensity score matching and difference-in-differences estimates of the causal effect of PE ownership. Together, these three stages provide an overview of the empirical results of PE operational performance in the Nordic context.

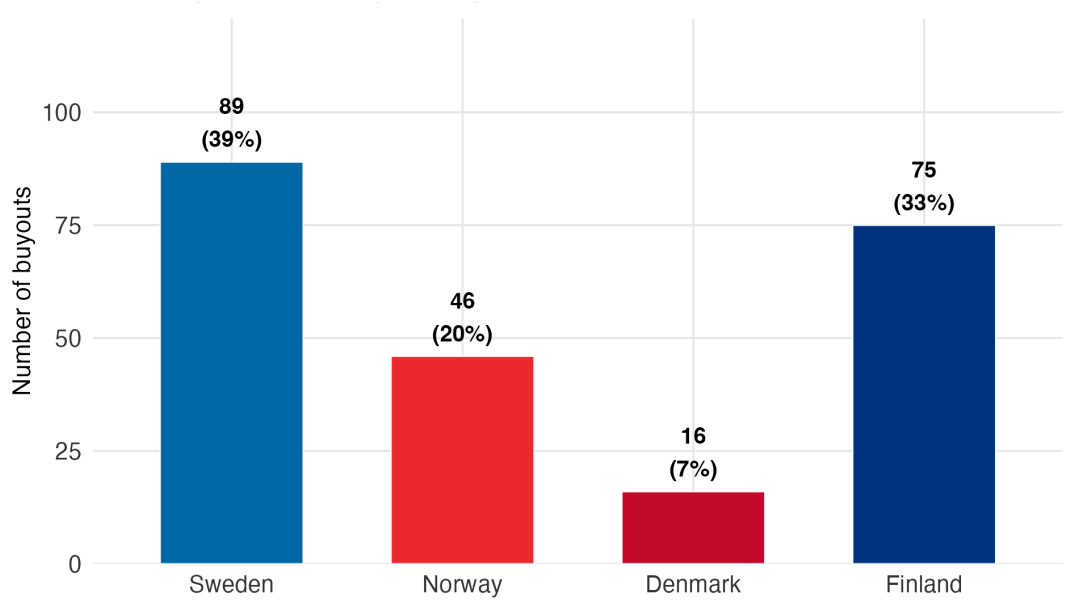
4.1 Sample Description and Univariate Analysis

4.1.1 Sample Characteristics

The final treatment sample consists of 226 completed Nordic PE buyouts spanning the period 1999 to 2023. As shown in Figure 1, Sweden constitutes the largest country sub-sample with 89 observations (39 percent of the total), followed by Finland with 75 observations (33 percent), Norway with 46 observations (20 percent), and Denmark with 16 observations (7 percent). The relative underrepresentation of Denmark reflects both the smaller scale of the Danish PE market relative to Sweden and Finland over the sample period and more limited Orbis coverage of Danish private transactions, and its consequences for the interpretation of the results are addressed in Section 7.8.

The Swedish dominance is compatible with the broader Nordic PE literature, which has consistently identified Sweden as the largest and most institutionally developed PE market in the region (Grubb & Jonsson, 2007; Gain.pro, 2025).

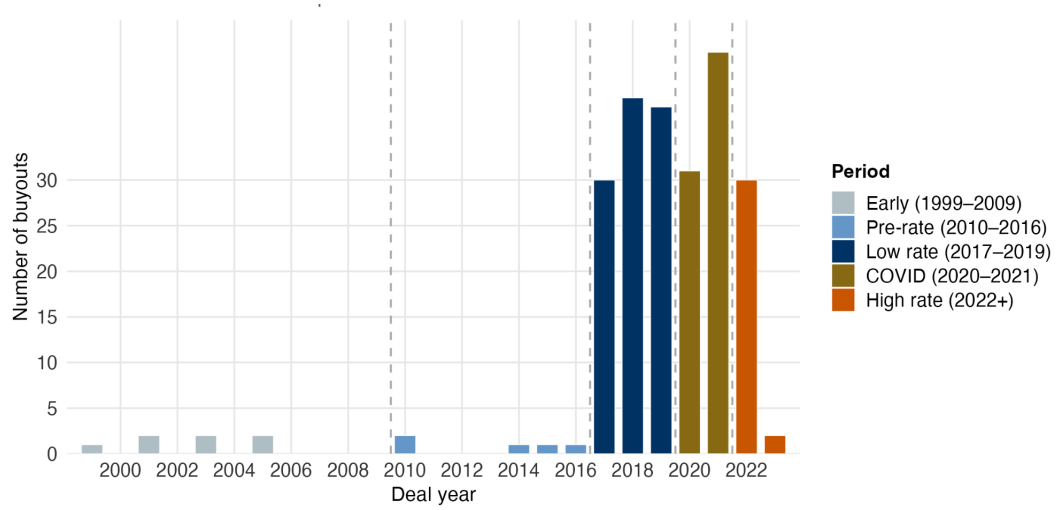
Figure 1: Distribution of Nordic PE Buyouts by Country



Note: Final sample: N = 226 completed buyouts, 1999 - 2023. Source: Bureau van Dijk Orbis M&A database. Sample restricted to completed IBO/MBO transactions with full ±2 year financial data. Deal year = year of completion.

Figure 2 illustrates the temporal distribution of buyouts across the five macroeconomic periods defined in Section 3. Deal activity is concentrated after 2016, with 107 transactions completed during the low-rate era of 2017 to 2019, 75 during the COVID period of 2020 to 2021, and 32 during the high-rate period from 2022 onwards. The early sample period from 1999 to 2009 contains only seven observations, and the pre-rate era from 2010 to 2016 contains five. This distribution reflects the well-documented surge in Nordic PE activity during the 2017 to 2022 period. The high-rate group of 32 observations, representing approximately 14 percent of the total sample, provides the basis for testing the macroeconomic moderation hypothesis (H2), though results should be interpreted with appropriate caution given the limited subsample size.

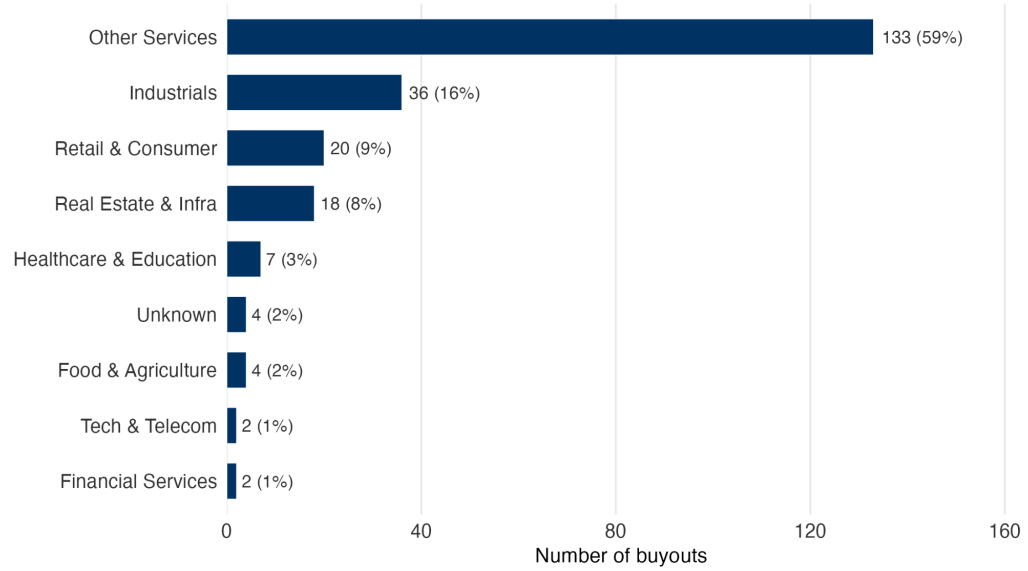
Figure 2: Annual Distribution of PE buyouts by Interest Rate Period



Note: Vertical dashed lines indicate period boundaries. Periods: Early = pre-2010, Low rate = 2017-2019, High rate = 2022+

The sector composition of the sample is presented in Figure 3. Other Services is the largest category with 133 observations (59 percent), followed by Industrials with 36 observations (16 percent), Retail and Consumer with 20 observations (9 percent), and Real Estate and Infrastructure with 18 observations (8 percent). This distribution is broadly in line with the Nordic PE market, where services and industrial companies are common targets for mid-market buyout activity. The high concentration in the Other Services category reflects both the heterogeneous nature of the NACE classification system and the service-intensive character of the Nordic economies. Sector fixed effects are included in all regression specifications, with Other Services used as the reference category, to control for systematic performance differences across industries.

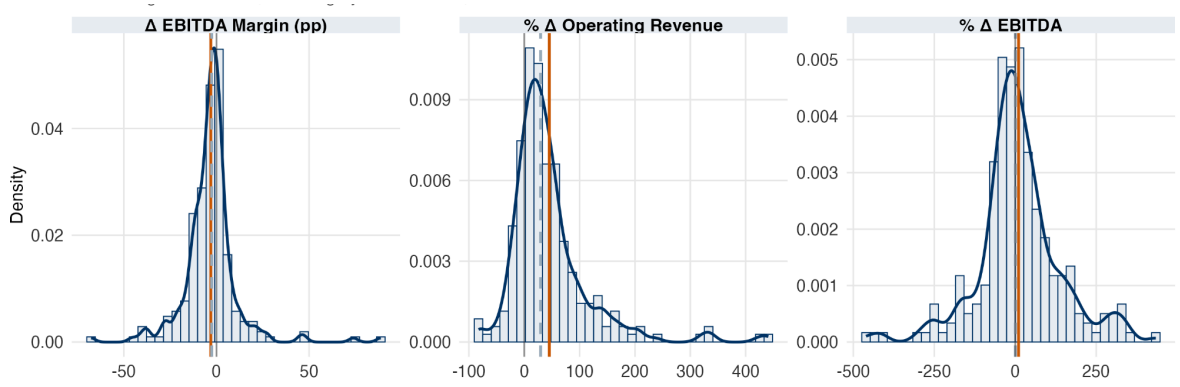
Figure 3: Sector Distribution of Nordic PE Buyouts



Note: N = 226 buyouts, sectors derived from Orbis NACE classification. Sector classification based on Orbis "Target major sector".

Table 2 shows descriptive statistics for the primary outcome variables and key deal characteristics. The pre-buyout EBITDA margin in the treatment sample has a mean of approximately 12.3 percent and a median of 11.3 percent. The difference between mean and median reflects the right-skewed distribution shown in Figure 4. The difference in means between PE targets and the control pool is not statistically significant ($p = 0.530$). This, together with the inclusion of initial EBITDA margin as a covariate in the OLS and PSM-DiD specifications, reduces the concern that the documented margin decline reflects mean reversion from unusually high pre-buyout profitability instead of PE-related effects. Institutional buyouts constitute approximately 96.5 percent of the sample, with the remaining 3.5 percent classified as management buyouts. The distributions of outcome variables, shown in Figure 4, are approximately symmetric for EBITDA margin change but markedly right-skewed for both revenue growth and EBITDA change, implying that a small number of high-growth observations pull the mean above the median for these variables.

Figure 4: Distribution of Outcome Variables



Note: Solid orange line = mean, dashed gray line = median, black line = zero reference. Winsorized at p1/p99 (i% ΔI > 500 excluded), N = 226 buyouts.

Table 2: Descriptive Statistics

Variable	N	Mean	Median	SD	p-value
Panel A: Outcome Variables (Treatment Sample, N = 226)					
ΔEBITDA Margin (pp)	226	-3.08	-2.6	14.8	0.002
%ΔOperating Revenue	226	45.3	29.5	72.4	< 0.001
%ΔEBITDA	226	10.4	0.48	130.32	0.234
Panel B: Deal Characteristics					
Pre-buyout EBITDA margin (%)	226	12.3	11.30	23.2	-
Log(Op.Revenue)t0	226	10.1	10.0	1.46	-
IBO (% of sample)	226	96.5%	-	-	-
High-rate period (2022+,%)	226	14.2%	-	-	-

Note: p-values for outcome variables are from two-sided paired t-tests. EBITDA margin change is winsorised at p1/p99. Revenue change excludes observations with |%Δ| > 500. SD = standard deviation. "-" indicates not applicable.

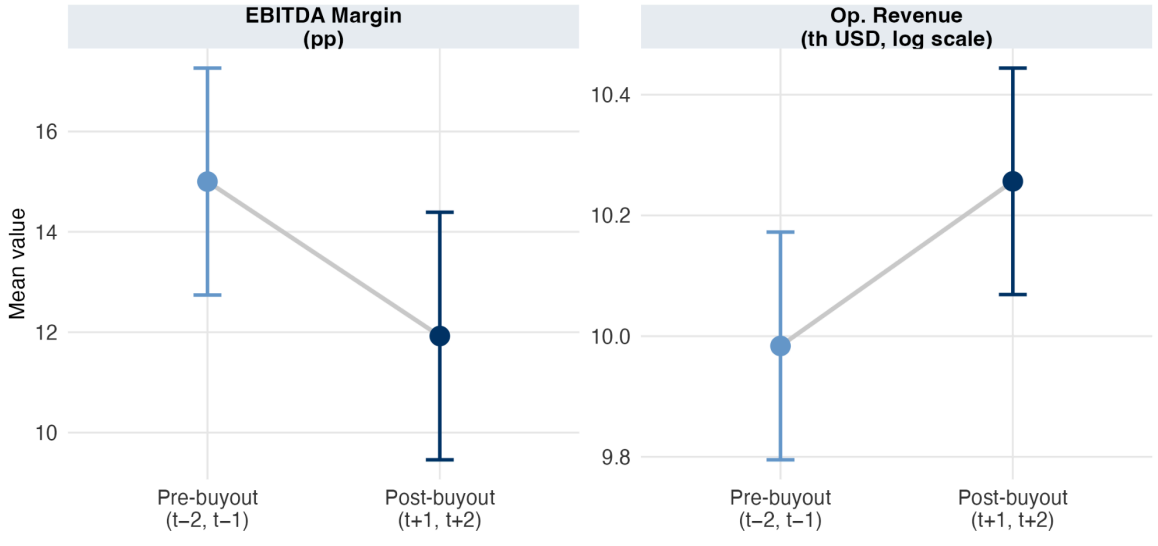
4.1.2 Before-After Univariate Analysis

The univariate before-after analysis tests whether the mean and median change in operational performance across the treatment sample are statistically different from zero. Figure 5 presents the mean pre- and post-buyout values of EBITDA margin and operating revenue, with 95 percent confidence intervals, for the full treatment sample.

The results reveal a clear contrast between the two primary outcome variables. For EBITDA margin, the mean declines from approximately 12.3 percentage points in the pre-buyout period (t-2, t-1) to approximately 9.2 percentage points in the post-buyout period (t+1, t+2), representing a mean change of -3.08 percentage points. Both paired t-test and Wilcoxon signed-rank test confirm that this decline is statistically significant at the one percent level ($p = 0.002$). The negative median change of approximately -2.6 percentage points, lower in absolute magnitude than the mean, reflects the presence of some large negative outliers in the tail of the distribution, as documented in Figure 4.

Operating revenue, by contrast, increases measurably over the same period. Mean post-buyout operating revenue is significantly higher than the pre-buyout level, with a mean percentage change of approximately 45 percent ($p < 0.001$) and a median change of approximately 30 percent. The log-scale representation in Figure 5 supports the direction and size of this increase while accounting for the skewness in revenue levels.

Figure 5: Pre- vs. Post-Buyout Operational Performance



Note: Mean with 95% confidence intervals. Pre = average of t-2, t-1. Post = Average of t+1, t+2. N = 226 Nordic PE buyouts. Operation revenue shown on log scale to reduce skew.

The combination of declining EBITDA margins and significant revenue growth is the central empirical finding of the univariate analysis. This pattern is consistent with PE sponsors in the Nordic context

prioritising top-line expansion over short-run improvement. Such a strategy may temporarily depress margins as firms invest in capacity, headcount, and market development before the revenue gains are fully realised.

The percentage change in EBITDA does not exhibit a statistically significant change in the post-buyout period (t-test $p = 0.234$, Wilcoxon $p = 0.335$), a result that is mechanically aligned with the simultaneous decline in margins and increase in revenues. Given the absence of a significant average effect and the high variance of this outcome variable (residual standard error exceeding 130 percentage points in the OLS specifications), the percentage change in EBITDA is not carried forward to the PSM-DiD analysis, where it would reduce estimation precision without adding analytical value.

These findings provide direct support for H1. The hypothesis predicted a statistically significant decline in EBITDA margin, consistent with the idea of the growth LBO mechanism, and the data show a significant decline of 3.08 percentage points ($p < 0.01$). This is in line with the expected direction of H1. The margin decline of 3.08 percentage points confirms that PE sponsors deploy investment-phase capacity building, including additional headcount, market entry expenditures, and management professionalisation, which reduce margins ahead of revenue realisation. The simultaneous revenue growth of 45 percent suggests that operational activity is expanding, and supports the interpretation that the margin decline reflects strategic investment as opposed to operational decline.

4.2 OLS Cross-Sectional Regression Results

To identify deal-level and macroeconomic factors associated with heterogeneity in post-buyout performance, OLS cross-sectional regressions are estimated for each of the three outcome variables. The baseline specification includes pre-buyout firm size (log operating revenue at t0), initial EBITDA margin, the IBO dummy, the low-rate and high-rate period dummies, and country and sector fixed effects. An extended specification adds pre-buyout leverage (debt-to-assets ratio, winsorized at p1/p99) as an additional control. All standard errors are HC1-robust. Full results for the EBITDA margin outcome and the revenue growth outcome are presented in Tables 3 and 4 respectively. Results for percentage change in EBITDA are reported in Table A.1 in the Appendix, given the absence of statistically significant coefficients on the variables of primary interest.

4.2.1 Determinants of EBITDA Margin Change (H1, H4)

Table 3 reports the OLS results for the change in EBITDA margin as the dependent variable. The most economically and statistically significant finding is the coefficient on log pre-buyout firm size, which is positive and significant at the one percent level across all three specifications. In the baseline model, shown in column 1, the coefficient is 1.685 (SE = 0.585, $p < 0.01$), implying that a one-unit increase in

log operating revenue at t0 corresponds to approximately 1.69 percentage points greater EBITDA margin improvement in the post-buyout period, after controlling for initial profitability, deal type, macroeconomic period, and country and sector fixed effects. This coefficient is stable across specifications, taking values of 1.592 (SE = 0.603) and 1.589 (SE = 0.601) in the leverage-augmented and low-rate-excluded models, respectively.

This finding is aligned with H4, which predicted that larger firms at the time of acquisition experience greater EBITDA margin improvement. This suggests that operational value creation may be more effective in firms with greater scale. This size effect is consistent with an economies-of-scale view of operational engineering (Kaplan & Strömberg, 2009). In the Nordic private-to-private context, where portfolio companies are predominantly small and medium-sized enterprises, the size effect likely reflects the increasing scope for professional management frameworks, cost rationalisation initiatives, and process optimisation as organisational complexity grows with firm size.

The IBO dummy is negative across all specifications, with a coefficient of approximately -7.9 percentage points in the baseline model (SE = 7.317), but does not reach conventional significance thresholds ($p > 0.10$ in all specifications). This indicates that, controlling for firm size and macroeconomic period, institutional buyouts do not generate systematically different EBITDA margin outcomes than management buyouts. The large standard errors on this coefficient reflect the substantial variation in margin outcomes within each group, in line with the wide distribution documented in Figure 4. The high-rate period dummy is negative but also falls short of significance in the margin regressions (-2.347, SE = 2.585 in the baseline), which suggests that the interest rate environment affects revenue growth more strongly than margin changes. This is theoretically possible, as higher financing costs directly constrain investment and expansion instead of necessarily reducing operational efficiency.

Taken together, the positive and statistically significant size coefficient across all three specifications supports H4: larger firms at the time of buyout experience greater post-buyout EBITDA margin improvement. This is supportive of the view that PE-led operational value creation may depend partly on firm scale in the Nordic context.

Table 3: OLS Regression Results - Δ EBITDA Margin (pp)

	(1) Baseline	(2) + Leverage	(3) No low-rate
Log(Op. Revenue)t0			
	1.685***	1.592***	1.589***
	(0.585)	(0.603)	(0.601)
IBO Dummy			
	-7.896	-7.501	-7.739
	(7.317)	(7.438)	(7.510)
High-rate era (≥2022)			
	-2.347	-2.473	-1.974
	(2.585)	(2.617)	(2.378)
Initial EBITDA Margin			
	-0.071	-0.064	-0.063
	(0.082)	(0.079)	(0.080)
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Observations	226	226	226
R^2	0.067	0.070	0.070
Adj. R^2	-0.004	-0.006	-0.002

*Note: Dependent variable: change in EBITDA margin (percentage points, post minus pre). HC1 robust standard errors in parentheses. Reference categories: Sweden (country), Other Services (sector). Leverage winsorised at p1/p99. * p < 0.10, ** p < 0.05, *** p < 0.01.*

4.2.2 Determinants of Revenue Growth (H2, H3)

Table 4 presents OLS results for the percentage change in operating revenue. Two findings stand out as statistically and economically significant: the high-rate period dummy and the IBO dummy. The

high-rate period dummy carries a coefficient of approximately -35 percentage points (SE = 11.200, $p < 0.01$) in the baseline specification, indicating that buyouts completed in 2022 or later are associated with revenue growth approximately 35 percentage points lower than otherwise comparable pre-2022 cohorts, after conditioning on firm size, initial profitability, deal type, and country and sector fixed effects. This coefficient remains stable when leverage controls are added: it takes a value of -35.002 (SE = 11.123) in the leverage-augmented specification, indicating that the result is not driven by omitted leverage variation.

This is the largest economic effect estimated in the OLS analysis, and it supports H2, though the estimate is sensitive to how the pre-2022 reference period is defined. The baseline reference category (deals outside the 2017-2019 and post-2021 windows) is dominated by the 2020-2021 cohort, which recorded the highest average revenue growth in the sample. When a separate 2020-2021 (COVID-era) indicator is added, so that the high-rate coefficient is estimated relative to pre-2017 deals, the coefficient falls to approximately -19 percentage points and is no longer statistically significant ($p = 0.26$), while the COVID-era coefficient is +18 percentage points ($p = 0.29$). The loss of significance partly reflects the small pre-2017 reference group (12 deals). Across specifications, including the subgroup DiD estimate in Section 6.2, the high-rate effect is therefore consistently large and negative (-18 to -35 percentage points), but the precise estimate and statistical significance depend on the comparison period and the small high-rate subsample. The result should accordingly be read as directionally robust but uncertain in size.

The scale of the high-rate effect is consistent with the theoretical channels outlined in Section 3: the sharp monetary tightening cycle initiated by Riksbanken, Norges Bank, and the European Central Bank from 2022 onwards increased debt servicing costs, reduced the cash available for operational investment and market expansion, weakened consumer demand, and tightened credit conditions across supply chains and customer bases. Figure 6, Panel B, illustrates this pattern. The high-rate cohort (2022+, n = 32) exhibits a mean revenue growth of approximately 15 percent, markedly lower than the 50 percent recorded for low-rate era deals (2017-2019, n = 107) and the 52 percent for COVID-era deals (2020-2021, n = 75). The low-rate dummy, by contrast, is positive but not significant (2.214, SE = 11.957), which supports the view that the low-rate environment from 2017 to 2019 was not related to systematically higher revenue growth relative to the pre-rate baseline, after controlling for deal-level characteristics.

The IBO dummy has a coefficient of 38.677 percentage points (SE = 19.129, $p < 0.05$) in the baseline revenue regression, indicating that institutional buyouts generate approximately 39 percentage points higher revenue growth than management buyouts, after controlling for size, initial profitability, macroeconomic period, and country and sector fixed effects. This coefficient is stable at 38.436 (SE =

19.101) in the leverage-augmented specification. The finding supports H3 and is aligned with the theoretical prediction that PE sponsors in institutional buyouts may bring stronger operational resources, greater strategic ambition, and a more growth-oriented ownership approach than management-led buyouts. As institutional buyouts make up 96.5 percent of the sample, this comparison relies on a small number of management buyouts and should be interpreted with caution, and is further discussed in Section 6.3.

The coefficient on log pre-buyout firm size in the revenue regressions is negative (-5.150, SE = 3.817 in the baseline) but does not reach conventional significance thresholds ($p > 0.10$). This suggests that pre-buyout size is not systematically aligned with revenue growth after controlling for the other variables, in contrast to its strong positive effect on EBITDA margin improvement. One possible interpretation is that smaller portfolio companies can achieve revenue growth comparable to larger firms under PE ownership, even if margin improvement from operational engineering depends more on firm size. Initial EBITDA margin does not significantly predict either outcome variable in the OLS specifications.

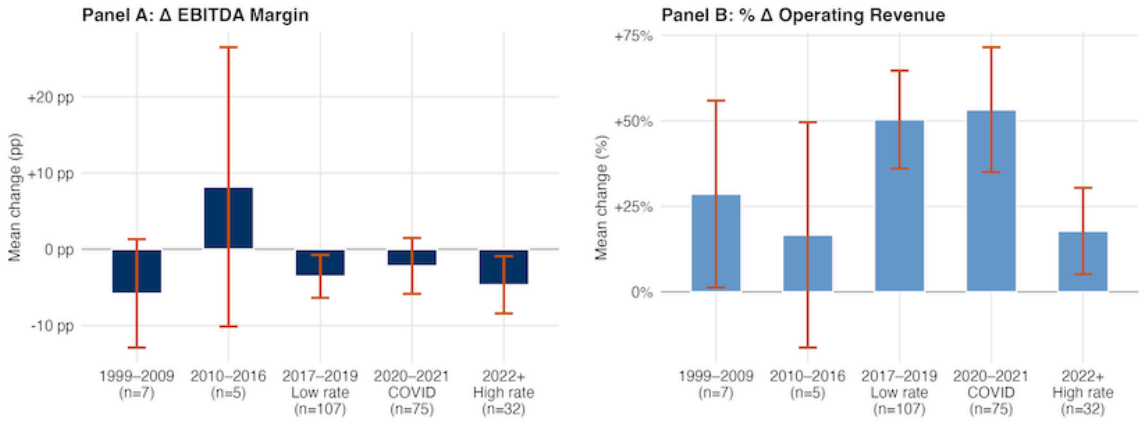
Table 4: OLS Regression Results - % Δ Operating Revenue

	(1) Baseline	(2) + Leverage
High-rate era (≥2022)		
	-35.079***	-35.002***
	(11.200)	(11.123)
IBO dummy		
	38.677**	38.436**
	(19.129)	(19.101)
Log(Op. Revenue)t0		
	-5.150	-5.093
	(3.817)	(3.990)
Initial EBITDA Margin		
	0.236	0.232

	(0.212)	(0.204)
Low-rate era (2017-2019)		
	2.214	2.167
	(11.957)	(12.261)
Country FE	Yes	Yes
Sector FE	Yes	Yes
Observations	226	226
R^2	0.081	0.081
Adj. R^2	0.010	0.005

*Note: Dependent variable: percentage change in operating revenue (post minus pre). HC1 robust standard errors in parentheses. Reference categories: Sweden (country), Other Services (sector). Leverage winsorised at p1/p99. * p < 0.10, ** p < 0.05, *** p < 0.01.*

Figure 6: Operational Performance by Deal Period



Note: Mean with 95% CI. Error bars = ±1.96 x SE. N per group shown in x-axis labels. "High rate" shows clearly lower revenue growth (consistent with OLS high_rate coefficient ≃ -35 pp).

The R^2 values across the OLS specifications are relatively low, ranging from 0.067 to 0.081. This indicates that the included regressors explain approximately 6-8 percent of the cross-sectional variation in post-buyout performance. This is compatible with the broader PE performance literature, where cross-sectional regressions on deal-level accounting outcomes routinely produce low explanatory power (Guo, Hotchkiss, and Song, 2011; Acharya et al., 2013). The low R^2 reflects the fact that post-buyout operational outcomes are highly firm-specific. Factors such as management quality,

competitive conditions, and strategic execution are not directly observed in accounting data, but likely account for much of the remaining outcome variation.

The low explanatory power has a specific implication for how the individual coefficients of interest should be interpreted. When a model explains only 6-8 percent of the cross-sectional variance, the estimated coefficients for the IBO dummy and log firm size are identified against substantial unexplained variation. This means that unobserved factors correlated with both deal type and operational outcomes, such as PE fund quality, sector specialization, or deal complexity, could partly explain the estimated associations. The IBO coefficient, in particular, may capture some variation in PE sponsor characteristics that is correlated with deal type but not separately controlled for. These coefficients should therefore be interpreted as conditional associations rather than structural effects, and their economic importance should be read with appropriate caution. The stability of both estimates across the baseline and leverage-augmented specifications provides some reassurance, but it does not remove this concern.

The primary objective of the OLS analysis is not prediction, but to estimate the specific coefficients of interest, namely the high-rate dummy, the IBO dummy, and log firm size. The statistical significance and stability of these coefficients across specifications is the relevant criteria for the inferences drawn in this study, with the limitation above in mind.

4.3 Propensity Score Matching and Difference-in-Differences Results

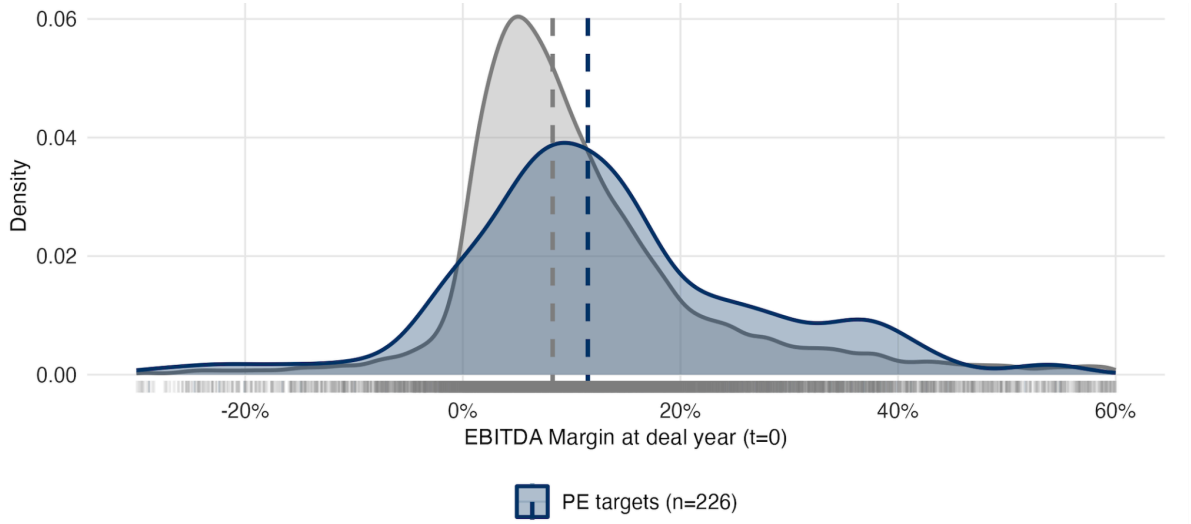
4.3.1 Matching Quality

Before presenting the DiD estimates, it is necessary to verify that the matching procedure achieves sufficient covariate balance between the treated firms and their matched controls. The standardised mean differences (SMD) for the matching covariates, reported in Table 8, improve after matching. Before matching, log operating revenue exhibits a substantial imbalance of approximately 0.25 SMD, with PE targets being somewhat smaller on average than the broad Orbis control pool, which contains larger, established firms. The propensity score model reduces this imbalance. Figure 7 shows a separate dimension of the raw difference between the groups: PE targets have a median pre-buyout EBITDA margin of 11.3 percent compared to 8.3 percent for the control pool, a gap of 3.0 percentage points between medians, although the difference in means is not statistically significant.

Following nearest-neighbour matching at a 3:1 ratio with a caliper of 0.20 standard deviations, the matching sample shows acceptable balance on the main matching variables. The post-matching SMD for log operating revenue is -0.120, slightly above the conventional 0.10 threshold recommended by Austin (2011), but this is directly controlled for in the DiD specification. The SMD for initial EBITDA

margin is -0.030, well within the acceptable range. Including country indicators in the propensity model ensures that the treated and control groups are balanced on country. No treated observations are unmatched, and the matching procedure produces 904 observations comprising 226 PE-backed firms and 678 matched controls, with one observation per firm reflecting pre-to-post first differences across the symmetric ±2-year event window. The parallel trends assumption is assessed by comparing pre-buyout EBITDA margin trends between treated and control firms. A t-test of pre-period trend differences gives p = 0.843, while a placebo regression using a fictitious treatment date in the pre-period gives an insignificant coefficient (p = 0.212). Both results are in accordance with the parallel trends assumption.

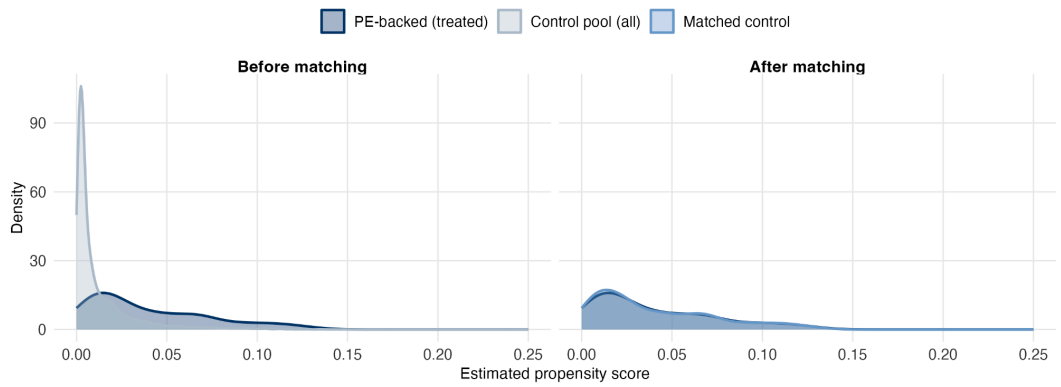
Figure 7: EBITDA Margin at t0: PE Targets vs. Control Pool



Note: Dashed lines = group medians. PE targets show modestly higher pre-buyout margins (difference not statistically significant). Motivates PSM matching on margin_t0. Median PE = 11.3% vs. control = 8.3% (+3.0 pp, mean difference not significant, p = 0.53)

Figure 8 presents the propensity score distributions before and after matching. Before matching, the control pool contains many firms with near-zero propensity scores, reflecting that PE buyouts are rare relative to the available firm universe. After 3:1 nearest-neighbor matching with a caliper of 0.20 standard deviations, the matched control distribution closely resembles that of the PE-backed firms. This suggests that common support holds across the effective propensity score range.

Figure 8: Propensity Score Overlap: Before and After PSM



Note: Left: Full control pool vs. PE targets. Right: PSM-matched sample. X-axis trimmed to [0,0.25], >99% of observations fall within this range. Propensity score estimated via logit: treated ~ log(op. revenue) + EBITDA margin + country FE. 3:1 nearest-neighbour matching, caliper = 0.20 SD.

4.3.2 Average Treatment Effect on the Treated

Table 5 reports the PSM-DiD estimates of the average treatment effect on the treated (ATT) for EBITDA margin change (column 1) and percentage change in operating revenue (column 2). The specification includes year, country, and sector fixed effects as well as PSM matching weights, and is estimated on 904 observations with standard errors clustered at the firm level.

The ATT for EBITDA margin is estimated at -2.618 percentage points (SE = 1.076, $p < 0.05$). This negative and statistically significant estimate suggests that PE-backed firms experience a margin decline of approximately 2.6 percentage points relative to their matched non-PE-backed controls, after accounting for time-invariant differences through the difference-in-differences structure. This finding fits with the before-after estimate of -3.08 percentage points from the univariate analysis, and suggests that the margin decline is less likely to be driven by broader economic trends or pre-existing firm characteristics alone.

The proximity of the DiD ATT estimate to the univariate before-after figure is itself informative. A large divergence between the two would suggest that economy-wide trends or systematic differences between PE targets and the broader firm population were driving the before-after result. The fact that the two estimates are relatively close (-2.618 pp vs. -3.08 pp), implies that macroeconomic trends affecting Nordic firms generally over the sample period did not substantially inflate the raw before-after comparison. This convergence supports the interpretation that the margin decline

documented in Section 4.1 reflects a PE ownership-related effect and not a coincidental overlap with broader industry or macroeconomic conditions.

The causal interpretation remains conditional on the parallel trends assumption and the observability of relevant confounders. Still, the DiD estimate is more informative than the before-after estimate because it accounts for how matched control firms develop over the same period.

The ATT for revenue growth is estimated at 14.331 percentage points (SE = 5.623, $p < 0.05$). This positive and significant estimate suggests that PE-backed firms generate approximately 14 percentage points more revenue growth than their matched non-PE-backed peers over the ±2-year event window. This provides evidence that PE ownership is associated with superior top-line performance beyond what would be expected from pre-existing firm characteristics. The magnitude of this effect is somewhat lower than the 45 percent mean growth documented in the univariate before-after analysis, which is expected. The DiD design subtracts the revenue growth of matched control firms over the same period, thereby accounting for broader economic and sector-specific growth trends that could otherwise inflate the before-after estimate.

Together, these findings support H5: relative to their matched non-PE-backed peers, PE-backed Nordic companies show significant revenue outperformance alongside margin compression, both significant at the five-percent level. The pattern of simultaneous margin compression and revenue outperformance in the DiD specification mirrors the pattern documented in the univariate analysis and supports the interpretation that PE sponsors in the Nordic market pursue a growth-over-margin operational strategy. The F-statistics for both DiD specifications are significant at the one percent level ($F = 2.117$ and $F = 1.848$, respectively), confirming the joint statistical significance of the specifications.

The log operating revenue control in the DiD revenue regression carries a coefficient of -6.439 (SE = 1.551, $p < 0.001$), indicating that within the matched sample, larger pre-buyout firms tend to have lower post-buyout revenue growth, compatible with a size-mean reversion effect whereby smaller firms grow proportionally faster from a lower base. This is a mechanical but important result, it confirms that the positive ATT on revenue is not driven by size imbalances between treated and control groups, since the matching procedure has already reduced this difference.

Table 5: PSM-DiD Estimates - Average Treatment Effect on the Treated

	(1) Δ EBITDA Margin (pp)	(2) % Δ Revenue
PE Treatment (ATT)		
	-2.618**	14.331**
	(1.076)	(5.623)
Log(Op. Revenue) t0		
	-0.177	-6.439***
	(0.262)	(1.551)
Initial EBITDA Margin		
	0.030	0.263
	(0.043)	(0.188)
Year FE	Yes	Yes
Country FE	Yes	Yes
Sector FE	Yes	Yes
PSM Matching Weights	Yes	Yes
Observations	904	904
R^2	0.066	0.058
F-stat	2.117***	1.848***

Note: Estimation on PSM-matched sample (N = 904, 226 PE-backed firms + 678 matched controls, first differences, one row per firm, 3:1 nearest-neighbour matching, caliper = 0.20 SD). Standard errors clustered at firm level. Propensity model: PE ~ log_size + margin_t0 + country.

4.4 Summary of Findings

Table 6 summarises the empirical support for each of the five hypotheses. H1 is supported. The hypothesis predicted a statistically significant decline in EBITDA margin in line with investment-phase

capacity building, and the data confirm a highly significant decline of 3.08 percentage points ($p < 0.01$). H2 is directionally supported: high-rate era buyouts show the largest negative revenue effect in the study (approximately 35 percentage points in the baseline OLS), but this estimate is sensitive to the reference period and is not robustly significant as noted in Section 4.2.2. H3 is directionally supported, with IBOs generating approximately 39 percentage points higher revenue growth than MBOs, though this rests on only eight management buyouts and should be read as suggestive. H4 is supported, as log pre-buyout firm size is positively and significantly associated with EBITDA margin improvement. H5 is supported, as the PSM-DiD ATT estimates show significant margin compression alongside revenue outperformance relative to matched controls. As detailed in the note below, the margin effect is robust to unobserved confounding while the revenue effect is directionally supported but more sensitive.

Table 6: Summary of Hypothesis Tests

Hypothesis	Prediction	Empirical Test	Key Result	Verdict
H1: Operational change	Statistically significant decline in EBITDA margin in the post-buyout period, consistent with investment-phase margin compression	Paired t-test and Wilcoxon signed-rank on Δ EBITDA margin	-3.08 pp, $p < 0.01$	**Supported**
H2: Interest rate environment	High-rate buyouts (2022+) show lower post-buyout revenue growth	High-rate dummy in OLS (% Δ operating revenue)	-35.1 pp, $p < 0.01$	**Directionally supported**
H3: IBO vs. MBO	IBOs generate higher post-buyout revenue growth than MBOs	IBO dummy in OLS (% Δ operating revenue)	+38.677 pp, $p < 0.05$	**Directionally supported**
H4: Pre-buyout firm size	Larger firms experience greater EBITDA margin improvement	Log-size coefficient in OLS (Δ EBITDA margin)	+1.69 pp per log unit, $p < 0.01$	**Supported**

	post-buyout			
H5: Causal PE effect	Relative to matched peers, PE-backed firms show revenue outperformance alongside margin compression	PSM-DiD ATT estimates (Δ EBITDA margin and % Δ revenue)	Margin: -2.62 pp ($p < 0.05$); Revenue: +14.3 pp ($p < 0.05$)	**Supported**

Note on H5: The margin ATT (-2.62 pp) is robust to unobserved confounding through $\Gamma = 1.5$ and represents a causally credible estimate. The revenue ATT (+14.3 pp) loses significance at $\Gamma = 1.1$ and should be treated as directionally supported but causally less robust. H5 is supported with this qualification.

This does not mean a simple failure of the hypothesis. Rather, it reflects the growth-over-margin pattern of Nordic PE sponsors documented throughout the analysis. The significant margin compression, shown in both the univariate tests and the PSM-DiD ATT estimate, still points to a measurable PE ownership effect.

The consistency of the growth-over-margin pattern across all three stages of the analysis, from the univariate before-after comparison through the OLS cross-sectional regressions to the PSM-DiD causal estimates, constitutes the central empirical contribution of this thesis. Nordic PE sponsors appear to deploy an operational strategy characterised by strong revenue expansion, tolerating short-run margin compression in pursuit of top-line growth and market position. This strategy appears constrained by the post-2021 interest rate environment, with deals completed from 2022 onwards showing the largest negative revenue effect in the study (approximately 35 percentage points in the baseline OLS). As discussed in Section 4.2.2, the scale and significance of this effect depend on the comparison period, so it is best read as suggestive of a structural shift rather than conclusive evidence of one.

5. Robustness Checks

The main PSM-DiD findings are tested through two robustness checks and two further analyses that examine the identification strategy. The first excludes buyouts completed before 2010, while the second tightens the propensity score caliper from 0.20 to 0.15 standard deviations. These two checks are applied to the PSM-DiD framework, as this is the primary causal identification strategy of the study. Third, the plausibility of the parallel trends assumption is assessed using pre-treatment data.

Lastly, a Rosenbaum sensitivity analysis evaluates how strong an unobserved confounder would need to be to overturn the estimated treatment effects. Together, the four checks address the four key threats to the DiD design: sample composition sensitivity, matching precision sensitivity, the plausibility of the identifying assumption, and robustness to unobserved confounding.

5.1 Exclusion of Pre-2010 Observations

The baseline sample includes seven buyouts completed between 1999 and 2009. This early cohort differs from the later sample in several ways: PE fund penetration was lower, secondary buyout activity was limited, and the macroeconomic context included both the dot-com downturn and the start of the global financial crisis. The control pools available for these early deal years are also small, because Orbis has more limited financial coverage of Nordic private companies in this period. This raises the concern that matched control firms for early-period deals may be less comparable than those used for post-2009 cohorts.

To assess sensitivity to the composition of the early-period sample, the full analysis is repeated after excluding all buyouts completed before 2010. This gives a restricted treatment sample of 219 observations. The PSM procedure is then re-estimated on this sample, and the DiD specification is applied to the resulting matched panel.

The results are reported in Table 8. The ATT for EBITDA margin change is -3.594 percentage points ($p < 0.01$) in the restricted sample, compared to -2.618 in the baseline. The ATT for revenue growth is +12.984 percentage points ($p < 0.05$), compared to +14.331 in the baseline. Both estimates remain statistically significant and point in the same direction as the baseline results. The slightly stronger margin ATT in the restricted sample is in line with the early-period deals coming from a noisier matching environment, where covariate balance was harder to achieve. Importantly, the growth-over-margin pattern is preserved. PE-backed firms continue to show margin compression together with higher revenue growth relative to matched controls. The main findings are therefore not driven by the small number of pre-2010 transactions.

5.2 Stricter Propensity Score Caliper

The baseline PSM procedure uses a caliper of 0.20 standard deviations of the estimated propensity score. To assess whether the ATT estimates are sensitive to matching precision, the procedure is repeated using a stricter caliper of 0.15 standard deviations. This requires matched pairs to be more similar on the observed covariates.

Under the 0.15 caliper specification, all 226 treated observations are successfully matched, producing results identical to the baseline: ATT for EBITDA margin -2.618 percentage points ($p < 0.05$) and ATT for revenue growth +14.331 percentage points ($p < 0.05$). The stricter caliper does not reduce the matched sample, suggesting that the control pool is dense enough for each PE-backed firm to be matched with at least three closely comparable control firms within 0.15 standard deviations of its propensity score. This is itself a positive indicator of matching quality, indicating that the 3:1 nearest-neighbor procedure produces matches that are already operating near the 0.15 tolerance. As a result, robustness check 2 adds no new variation relative to the baseline specification, and the two sets of estimates remain unchanged.

This outcome implies that the estimated propensity scores for treated and control firms show strong common support across the full distribution. The overlap condition, which is a necessary assumption for a valid PSM inference is therefore satisfied not only at the conventional 0.20 caliper but at the stricter 0.15 threshold, providing additional confidence that the matching procedure identifies credible comparable control firms rather than borderline matches.

One matching quality concern that applies to both baseline and robustness check 2 is the standardised mean difference for log operating revenue, which stands at -0.120, slightly above the conventional 0.10 threshold recommended by Austin (2011). The SMD for initial EBITDA margin is -0.030, well within the acceptable range. The log size imbalance reflects that the broad Orbis control pool contains larger firms on average than the mid-market PE targets in the sample. The propensity score model reduces, but does not fully remove this imbalance. The remaining size imbalance is controlled for directly in the DiD specification through the log operating revenue covariate, which has a coefficient of -6.439 ($p < 0.001$) in the revenue regression. This confirms that size differences between matched pairs do not drive the ATT estimates. The robustness check 1 specification achieves stronger balance, with post-matching SMDs of -0.013 for log size and 0.044 for initial margin, both comfortably below 0.10.

5.3 Parallel Trends Validation

The validity of the DiD estimator relies on the parallel trends assumption. In the absence of PE ownership, treated and control firms would have followed the same trajectory in EBITDA margin and operating revenue. Although this assumption cannot be tested directly after treatment, its plausibility can be examined using pre-treatment data.

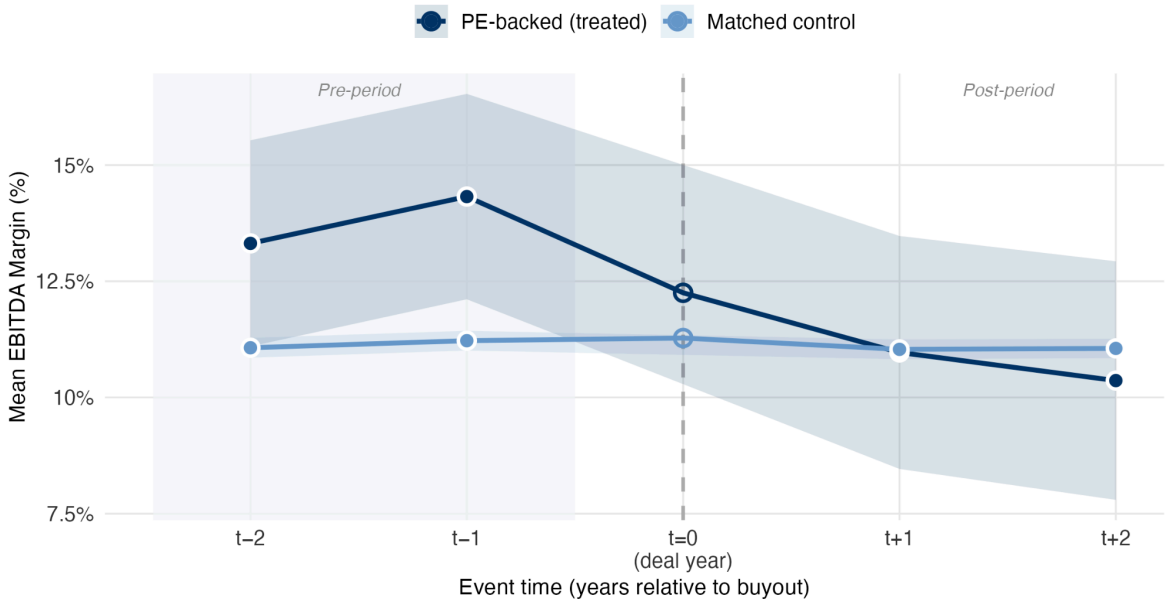
Two complementary tests are conducted. First, a two-sample t-test compares pre-period EBITDA margin trends (Δt-2 to t-1) between the matched treatment and control groups. The test gives a

p-value of 0.843, which suggests that there are no statistically significant differences in pre-period margin changes between the two groups. This supports the possibility of the parallel trends assumption for the primary outcome variable.

Second, a placebo DiD is estimated by shifting the treatment indicator one period earlier. The "pseudo-treatment" is assigned at t-1 instead of t=0, and the outcome is the pre-period change from t-2 to t-1. Under the null hypothesis of parallel pre-trends, this specification should produce an ATT estimate not statistically different from zero. The placebo ATT is not statistically significant (p=0.212), confirming the absence of a spurious pre-treatment divergence.

Figure 9 shows the pre- and post-period margin trajectories for matched treatment and control firms. The pre-period lines run closely parallel, consistent with both formal tests, while the post-treatment divergence is clearly visible from t=0 onwards. Taken together, the t-test, placebo DiD, and graphical inspection point in the same direction. The parallel trends assumption is supported by the available pre-period data, and the post-buyout changes identified by the DiD estimator are not attributable to pre-existing trend differences between the two groups.

Figure 9: Parallel Pre-trends: EBITDA Margin Trajectory



Note: Mean EBITDA margin (±95% CI) by event time. Pre-period trend test: p=0.843 (t-test). Placebo DiD: p=0.212. PE Sample: N = 226 buyouts (completed deals). Control: available firms from Orbis control pool matched to deal years. t=0 = deal year (open circle), excluded from pre/post averages. Shaded area = 95% confidence intervals.

5.4 Sensitivity to Unobserved Confounding

A further robustness check addresses the conditional independence assumption directly. While PSM removes selection bias attributable to observable firm characteristics, unobservable factors such as hidden growth potential or management quality may still explain part of the estimated treatment effects. To assess the severity of this concern, a Rosenbaum sensitivity analysis is conducted using 226 matched pairs extracted from the 3:1 PSM sample. For each matched subclass, the nearest available matched control firm is retained.

The analysis evaluates the strength an unobserved confounder would need to have, measured by Γ, the odds ratio of differential treatment assignment, to render the ATT estimates insignificant. The results are reported in Table 7.

Table 7: Matched Pair for Rosenbaum-Test

Γ	Rev p_lower	Rev p_upper	Margin p_lower	Margin p_upper
1.0	0.018	0.018	0.000	0.000
1.2	0.001	0.181	0.000	0.001
1.4	0.000	0.533	0.000	0.013
1.6	0.000	0.827	0.000	0.081
1.8	0.000	0.956	0.000	0.251
2.0	0.000	0.992	0.000	0.490

Note: Rosenbaum sensitivity bounds based on 226 matched pairs (1:1 nearest control). Γ = odds ratio of differential PE selection due to unobserved confounder. p_upper = conservative (worst-case) p-value. Revenue ATT tested against H0: no positive effect. Margin ATT tested against H0: no negative effect.

The margin ATT is highly robust to unobserved confounding. The worst-case p-value (p_upper) remains below 0.05 through Γ = 1.5 and only reaches 0.49 at Γ = 2.0, indicating that an unobserved confounder would need to increase the odds by about 50 percent of PE selection to invalidate the margin finding. Overall, this provides strong support that the margin compression documented in Section 4 is more consistent with a genuine ownership effect than selection bias.

The revenue ATT is clearly more sensitive. Significance is lost at Γ = 1.1, meaning a confounder that increases PE selection odds by as little as 10 percent could in principle, explain away the entire positive revenue effect. This threshold should be interpreted in light of the PE due diligence process, where sponsors often target firms with above-average growth potential, strong management teams, and identifiable growth opportunities. These factors are not captured by the observable matching covariates of firm size, initial margin, and country of origin. It is therefore possible that matched control firms differ from PE targets along dimensions that could produce a Γ in the range of 1.2 to 1.4. The revenue ATT of +14.3 percentage points should therefore not be interpreted as a precise causal estimate. It is best interpreted as an upper bound on the causal PE revenue effect, with the true effect potentially being considerably smaller or, in an extreme scenario, indistinguishable from zero after accounting for unobserved selection. The directional conclusion that PE ownership is associated with superior revenue performance remains credible and is supported by the OLS evidence in Section 4.2, but the causal effect is uncertain.

Taken together, the Rosenbaum analysis strengthens confidence in the margin finding as a causal effect of PE ownership, while indicating that the revenue ATT should be interpreted with greater caution.

5.5 Summary

Table 8 summarises the ATT estimates across both specifications. The sign, magnitude, and statistical significance of both outcome variables are stable across the two substantive robustness checks, with ATT estimates for EBITDA margin ranging from -2.618 to -3.594 percentage points and for revenue growth from +12.984 to +14.331 percentage points, all significant at the five percent level or better. The parallel trends validation also supports the causal interpretation of these estimates. Overall, the results suggest that the growth-over-margin pattern identified in Section 4 is not driven by sample composition or the choice of matching specification, but it appears to be a stable feature of the Nordic PE sample.

Table 8: Robustness checks - PSM-DiD average treatment effect on the treated

	Δ EBITDA Margin (pp)		**%Δ Operating Revenue**	
	(1) Baseline	**(2) R1: Post-2009**	**(3) Baseline**	**(4) R1: Post-2009**
PE treatment (ATT)	-2.618**	-3.594***	+14.331**	+12.984**

	(1.076)	(1.167)	(5.623)	(5.670)
EBITDA margin t0	0.030	0.028	0.263	0.241
	(0.043)	(0.044)	(0.188)	(0.178)
Log(op. revenue) t0	-0.177	-0.162	-6.439***	-6.201***
	(0.262)	(0.271)	(1.551)	(1.498)
Model diagnostics				
Year / Country / Sector FE	Yes	Yes	Yes	Yes
PSM weights	Yes	Yes	Yes	Yes
Caliper (SD)	0.20	0.20	0.20	0.20
PE deals matched (n)	226	219	226	219
Unmatched PE deals	0	0	0	0
Observations	904	876	904	876
Covariate balance (post-matching SMD)				
SMD log size	-0.120Δ	-0.013	-0.120Δ	-0.013
SMD margin t0	-0.030	0.044	-0.030	0.044

*Note: Estimation on PSM-matched sample (3:1 nearest-neighbor, caliper as stated). Standard errors clustered at the firm level in parentheses. Year, country, and sector fixed effects included in all specifications. ATT = average treatment effect on the treated. SMD threshold: < 0.10 = sufficient balance (Austin, 2011), Δ = marginally above threshold, controlled via regression covariate. Robustness Check 2 (caliper = 0.15) is not reported separately as it produces results identical to baseline. All 226 PE deals were matched within the tighter tolerance, confirming dense control pool coverage. * p < 0.10, ** p < 0.05, *** p < 0.01.*

Table 9: Summary of robustness checks

ATT estimates across specifications. All specifications include year, country, and sector fixed effects with PSM matching weights.

Specification	N(PE)	Caliper	ATT Δ margin	ATT %Δ rev	Verdict
Baseline	226	0.20	-2.618**	+14.331**	Stable
R1: Exclude pre-2010	219	0.20	-3.594***	+12.984**	Stable
R1 achieves better size balance (SMD -0.013 vs. -0.120 in baseline) and slightly strengthens the margin ATT.					
R2: Caliper 0.15	226	0.15	-2.618**	+14.331**	Identical
R2 is identical to baseline. Tighter caliper did not exclude any PE deals, confirming dense control pool coverage.					

*Note: * p < 0.10, ** p < 0.05, *** p < 0.01. Standard errors clustered at the firm level. Margin ATT in percentage points. Revenue ATT in percent.*

6. Discussion

6.1 The Growth-Over-Margin Strategy: Reconciling the Findings with Existing Theory

The main empirical finding of this thesis is that Nordic PE buyouts are associated with significant revenue growth while EBITDA margins decline. This pattern appears across all three parts of the analysis: the univariate before-after comparison, the OLS cross-sectional regressions, and the PSM-DiD causal estimates. The ATT estimates of -2.618 percentage points for EBITDA margin and +14.331 percentage points for revenue growth, both statistically significant at the five percent level, are in accordance with the descriptive evidence. The robustness checks also show that this pattern is not sensitive to sample composition or matching specification, which makes the overall interpretation more credible.

The negative margin finding may at first seem to conflict with Jensen's (1986) agency cost framework, which predicts that PE ownership through the disciplinary use of debt, concentrated ownership, and high-powered incentives should improve operational efficiency and thereby expand margins. The early empirical work by Kaplan (1989) and Smith (1990), which documented significant operating

cash flow improvements and efficiency gains in U.S. management buyouts of the 1980s, provided strong support to this prediction. However, the context of those findings was a wave of transactions targeting persistently over-diversified and inefficient public corporations, where the primary source of value was the elimination of agency costs instead of organic growth. The Nordic market of the 1999 to 2023 period presents a fundamentally different target profile, predominantly private-to-private transactions involving small and medium-sized enterprises that are typically well-managed, family-owned, or founder-led businesses with limited prior exposure to institutional capital.

In this context, the margin compression documented in the present study fits more naturally with what Boucly, Sraer, and Thesmar (2011) term the "growth LBO" hypothesis. Studying 839 French buyouts, Boucly et al. show that buyouts ease the credit constraints facing financially dependent firms, enabling investment programs that reduce short-run margins while allowing longer-run revenue growth. The mechanism is not cost-cutting efficiency but growth-enabling capital deployment: PE sponsors inject financial and managerial resources that allow portfolio companies to hire, expand into new markets, and invest in organisational capacity ahead of revenue realisation. The simultaneous decline in margins and increase in revenues observed in the Nordic sample is in accordance with this pattern, as investment-phase costs including additional headcount, market entry expenditures, and management professionalisation, are reflected as margin compression in the ±2-year observation window before the associated revenue gains are fully realised.

This interpretation is also supported by the auxiliary labor productivity analysis reported in Appendix Table A.4, and described in Section 3.4. Among the 219 buyouts with sufficient employee records in Orbis, revenue per employee increases by an average of 12.1 percent in the post-buyout period (median: 2.8 percent, t-test $p < 0.001$), while headcount grows by an average of 39.6 percent against a mean revenue growth of 45.3 percent. The key pattern is that employment expands nearly as fast as revenue, leaving only a limited gain in output per worker. This is difficult to fit with a pure efficiency-based interpretation, under which output per worker would be expected to rise much faster than total revenue as the workforce is rationalised. Instead, the small positive productivity increase points toward capacity-building, where PE sponsors invest in headcount to support expansion and realise only limited near-term productivity gains while future scale benefits are still developing.

A further alternative explanation for the observed revenue outperformance should also be considered. The Nordic PE market is characterised by high add-on activity, given the high B&B intensity of the Nordic market. A meaningful share of the revenue growth documented in this study may therefore reflect inorganic acquisition activity rather than organic operational improvement. The

present data do not allow for a clear separation of these two sources of growth, and the revenue ATT of +14.3 percentage points should be interpreted with this in mind. To the extent that buy-and-build execution itself constitutes a PE-specific operational capability, this does not necessarily weaken the conclusion that PE ownership drives superior revenue performance, but it does qualify the interpretation of the underlying mechanism.

This interpretation is also supported by the StepStone Group's (2023) value-creation decomposition for Nordic PE, in which revenue growth, as opposed to margin expansion, accounts for the largest share of value added. The present study's findings therefore align well with the existing Nordic-specific evidence, while adding a causal dimension through the PSM-DiD design that prior studies have not included.

The comparison with Thomsen and Christoffersen (2024) is useful, as their study of Nordic PE buyouts identifies a similar top-line versus bottom-line divergence. Their findings that PE ownership significantly boosts turnover growth but does not consistently translate into improved EBIT margin or ROA in the first half of the holding period is closely in line with the results of this study. The fact that two Nordic studies, based on different samples, time periods, and methodologies, identify a similar pattern makes it less likely that the growth-over-margin is driven only by sample-specific or methodological factors. Instead, it points to a recurring growth-over-margin pattern among Nordic PE sponsors. Both studies also raise the possibility that margin effects may change over longer holding periods, as investment-phase costs are absorbed and revenue growth increases. This is a question that the ±2-year window used in the present study cannot answer definitively.

This connects to the meta-analytic finding of Verbouw et al. (2025) that institutional quality shapes post-buyout outcomes: well-governed, investor-friendly markets are associated with larger PE-induced improvements. The Nordic countries, consistently ranked among the world's most transparent and institutionally developed markets, represent precisely the kind of environment where PE operational engineering should be expected to work well. However, the evidence here suggests that the value creation in this environment comes mainly through growth, and not efficiency gains. This is a distinction that may be less visible in aggregate meta-analytic evidence. The finding therefore suggests that the institutional context hypothesis should be interpreted more carefully, better institutions may not only increase the amount of PE value creation, but also influence how that value is created.

In countries with strong investor protection and well-developed equity markets, the growth channel dominates, while efficiency extraction is less central. The Nordic institutional environment,

characterised by high levels of investor protection, transparent governance frameworks, and deep capital markets, therefore does not simply provide favorable conditions for PE activity in the abstract. It specifically encourages the growth-oriented ownership model that the present findings document. This suggests that the growth-over-margin pattern identified in this thesis is not independent of the Nordic context but is, at least in part, a product of it. Future research examining PE operational performance in more stakeholder-oriented or less institutionally developed markets may find a qualitatively different pattern, with efficiency extraction playing a more prominent role relative to revenue growth.

The Rosenbaum sensitivity analysis reported in Section 5.4 adds an important qualification to the causal interpretation of the revenue ATT. The margin ATT is less sensitive to unobserved confounding, remaining significant through $\Gamma = 1.5$. This means that an unobserved confounder would need to increase the odds of PE selection by about 50 percent before the margin loses statistical significance. The revenue ATT, however, loses significance at $\Gamma = 1.1$, implying that a confounder increasing PE selection odds by as little as 10 percent could in principle explain away the positive revenue effect.

A further qualification to this overall interpretation follows from the buy-and-build decomposition described later in Section 6.4. The combination of margin compression and revenue growth that characterises the growth-over-margin pattern is not observed within either value-creation profile separately. It appears only when the two groups are pooled. Standalone investments drive the margin compression, but show no significant revenue effect. Buy-and-build platforms drive the revenue growth, but show no significant margin effect. The aggregate result therefore partly reflects the structure of two different strategies, instead of one common operational pattern across all portfolio companies.

This qualification refines the interpretation, but does not overturn it. At the aggregate level, Nordic PE sponsors appear to prioritise revenue growth while accepting short-run margin pressure, consistent with the growth-LBO hypothesis. However, the underlying mechanisms differ. For standalone investments, margin compression is most likely linked to investment-phase costs during organic growth. For B&B platforms, revenue growth is likely to reflect some consolidation of acquired revenues, and not purely organic expansion. As shown in Section 6.4, this contrast also helps explain the asymmetric robustness in Section 5.4. The margin result is driven by the larger standalone group and remains robust to substantial unobserved confounding. The revenue result relies more on the smaller B&B group and is therefore more sensitive. Margin compression is therefore the more robust and more general ownership effect, while revenue outperformance, although economically meaningful, is more heterogeneous and less secure as a causal claim.

The directional conclusion is supported by the convergent OLS evidence in Section 4.2, where the IBO dummy and high-rate coefficient confirm the direction and heterogeneity of revenue effects across an entirely different identification strategy, while the precise magnitude of +14.3 percentage points should not be treated as a stable causal estimate.

6.2 Macroeconomic Regime as a Moderator: The Interest Rate Channel

The finding that buyouts completed in 2022 or later are connected to approximately 35 percentage points lower post-buyout revenue growth is the largest effect estimated in the study and constitutes a contribution to the PE operational performance literature, although its precise effect and significance are sensitive to the comparison period. While the coefficient is significant at the one percent level in the baseline and leverage-augmented OLS specifications, it is not robust to separating the COVID-era cohort from the reference group, and its magnitude depends heavily on the choice of comparison period. The result is therefore best read as suggestive, and not conclusive.

The theoretical channels through which rising interest rates constrain post-buyout revenue growth are several and often reinforce each other. Axelson, Jenkinson, Strömberg, and Weisbach (2013) provide the foundational evidence that buyout leverage is primarily driven by broader credit conditions, not company-specific fundamentals. This implies that buyouts financed when credit was cheap may end up with capital structures that grow considerably more burdensome once monetary conditions tighten. For deals completed in 2022 or later, after Riksbanken, Norges Bank, and the European Central Bank had started a sharp monetary tightening cycle, the direct cost-of-capital effect is combined with broader macroeconomic pressures.

Beyond the direct cost of debt, higher rates weaken end-customer demand and reduce the willingness to pursue add-on acquisitions that support buy-and-build growth, compounding the pressure on top-line performance. Nordic deal volumes fell approximately 40 percent from their 2021 peak (Gain.pro, 2025), and the broader slowdown in PE activity through 2023 documented by the KPMG Nordic PE market update (2025) is in line with the estimated high-rate coefficient.

The subgroup DiD analysis described in Section 3.9 adds further support for this interpretation. When the matched PSM sample is restricted to pre-2022 buyouts (n = 194), the revenue ATT is +17.66 percentage points (p = 0.008), broadly consistent with the full-sample baseline. For high-rate era deals (n = 32), the revenue ATT turns negative to -17.7 percentage points, though this estimate does not achieve conventional significance (p = 0.257). As noted above, the high-rate subsample has only around 25 to 30 percent power to detect an effect of the full-sample magnitude, and the 95 percent confidence interval for this estimate spans from approximately -48 to +13 percentage points. The lack

of statistical significance therefore appears to reflect limited statistical power more than clear evidence of no effect. The margin ATT for the high-rate subgroup is larger in absolute terms than in the pre-2022 cohort (-5.6 vs. -2.8 percentage points) and is marginally significant (p = 0.086), which could indicate that the margin compression may become stronger under tighter financing conditions, even as the revenue effect weakens. Given the small high-rate subsample, this estimate should also be interpreted as suggestive instead of conclusive. The reversal in revenue growth, from a significant effect in the pre-2022 cohort to a large but imprecisely estimated negative point estimate in the high-rate cohort, supports the interpretation of the macroeconomic moderation hypothesis and is in line with the OLS results, although the causal magnitude for the high-rate period cannot be pinned down without a larger sample. These results are reported in Table 10.

Table 10: Subgroup DiD Results

Subgroup	N (PE)	ATT Δ Margin	p	ATT % Δ Revenue	p
Low/mid-rate (<2022)	194	-2.81**	0.025	+17.66***	0.008
High-rate (≥2022)	32	-5.60*	0.086	-17.70	0.257
IBO	218	-2.75**	0.012	+16.24***	0.004

*Note: Subgroup DiD estimated on PSM-matched subclasses. Clustered standard errors at the firm level. Year, country, and sector FE included. *p<0.10, **p<0.05, ***<0.01.*

What is worth noting is that the high-rate dummy does not reach conventional significance in the EBITDA margin regressions (coefficient -2.347, p > 0.10). This difference between the margin and revenue results is compatible with the underlying theory: rising financing costs may limit investment and expansion capacity, which affects revenue growth more directly than the operational efficiency of existing business activities. A PE-backed firm facing higher debt servicing costs may reduce its market expansion program but maintain or even improve the margin of its existing operations through cost discipline. This suggests that the interest rate channel operates primarily through the investment and growth capacity of portfolio companies, more than their underlying operational efficiency. For PE practitioners, this means that the post-2021 period should mainly be understood as a constraint on growth capacity, not necessarily as evidence of weaker operating efficiency.

This result also raises a broader point for the PE literature. Most existing academic studies of PE operational performance, including the U.S. evidence in Kaplan (1989), Guo, Hotchkiss, and Song

(2011), and Cohn, Mills, and Towery (2014), as well as the European evidence in Boucly et al. (2011) and Acharya et al. (2013), were conducted during, or mainly draw on periods of low or neutral interest rates. The post-2021 tightening cycle provides a useful test of whether PE operational value creation holds under different financing conditions. The present evidence suggests it does not hold in the same form: the growth-over-margin strategy that characterises Nordic PE in the low-rate era appears to be constrained when the cost of capital rises sharply. Whether this reflects a temporary cyclical disruption or a more lasting change in PE operational outcomes remains an open question. Future research with a longer observation window, including 2025 and later years, could help address this question.

6.3 Deal Type and Firm Size Heterogeneity

The finding that institutional buyouts are associated with approximately 39 percentage points higher post-buyout revenue growth than management buyouts, after controlling for firm size, initial profitability, macroeconomic period, and country and sector effects, reflects the theoretical prediction that PE sponsors in IBOs bring stronger operational resources and a more active growth agenda than what is typically available in MBOs.

Acharya, Gottschalg, Hahn, and Kehoe (2013) provide relevant empirical support for this interpretation. They find that PE partners with an operational background are associated with stronger operational improvements, which they link to the operational resources, management frameworks, and strategic networks that experienced sponsors bring to portfolio companies. In the Nordic context, where secondary buyouts constitute a significant share of IBOs, the acquiring sponsor often has sector knowledge built through previous platform investments. MBOs, by contrast, preserve strategic continuity and align managerial incentives, as emphasised by Jensen (1986, 1989), but may offer less room for broader strategic repositioning.

This estimate should be interpreted with some caution, since the sample contains only eight management buyouts and institutional buyouts account for 96.5 percent of observations. The deal-type difference should therefore be viewed as suggestive.

It is also worth noting that the IBO dummy does not reach significance in the EBITDA margin regression (coefficient -7.896, $p > 0.10$). This suggests that although IBOs are associated with stronger revenue growth, EBITDA margin changes under IBO and MBO ownership are not systematically different after controlling for firm size. One possible explanation is that both deal types involve costs during the investment phase. For IBOs, these costs may come from active growth initiatives, while for MBOs, they may come from incentive restructuring costs. The difference is that the stronger growth

ambition in IBOs appears to translate mainly into higher revenue growth over the two-year post-buyout window, rather than into systematically different margin outcomes.

The finding that larger firms at the time of acquisition experience significantly greater EBITDA margin improvement (+1.685 percentage points per log unit of revenue, $p < 0.01$) fits with an economies-of-scale view of operational engineering (Kaplan & Strömberg, 2009), in which greater organisational complexity gives sponsors more room to add value. In the Nordic private-to-private context, this likely reflects the greater availability of clear improvement opportunities, such as professional management frameworks, cost rationalisation, and process standardisation as organisational complexity grows with firm size. Very small targets, by contrast, may primarily require capacity investment before revenue gains are realised, which puts downward pressure on margins in the short run. The lack of a significant size effect on revenue growth (coefficient -5.150, $p > 0.10$) suggests that this pattern is specific to margin improvement, not a broader size advantage in PE value creation. Smaller Nordic portfolio companies appear equally able to generate revenue growth under PE ownership.

6.4 Buy-and-Build Strategy and Operational Performance

This section presents exploratory evidence on whether PE-backed companies pursuing a buy-and-build (B&B) strategy exhibit different operational trajectories than standalone PE investments. Of the 226 matched platform companies, 77 (34.1%) are identified as B&B acquirors based on the Orbis M&A cross-match described in Section 3.3. Table 11 presents mean outcomes for B&B and non-B&B platforms within the PE group.

Table 11. Buy-and-Build vs. Standalone PE: Descriptive Comparison (PE Group Only)

Group	n	ΔEBITDA Margin (pp)	SD	%Δ Operating Revenue	SD
Non-B&B (standalone)	149	-4.50	15.05	+37.2%	62.3
B&B (≥1 add-on)	77	-0.32	14.14	+61.0%	87.1
Difference (B&B vs. Non-B&B)	-	+4.19 pp**	-	+23.79 pp**	-
p-value (Welch t-test)	-	0.041	-	0.035	-

*Note: ΔEBITDA Margin = post-period mean minus pre-period mean (pp). %Δ Operating Revenue = percentage change in operating revenue. Significance: ** p < 0.05.*

B&B platforms experience limited margin compression compared to standalone PE investments (-0.32 pp vs. -4.50 pp) and a significantly higher revenue growth (+61.0% vs. 37.2%), with both differences statistically significant at the 5% level.

After controlling for pre-deal firm characteristics, deal year, country, and sector fixed effects (n = 226), the B&B dummy remains statistically insignificant for margin (β = +3.45 pp, SE = 2.62, p = 0.189) but remains positively associated with revenue growth (β = +25.3 pp, SE = 12.3, p = 0.042). The add-on count, used as an intensity measure, is not statistically significant in either model (margin: β = +0.21, p = 0.240, revenue: β = +0.71, p = 0.714). This suggests that B&B status itself, instead of the number of add-on acquisitions, is related to the revenue effect. This revenue advantage, together with the absence of an add-on-count effect, is consistent with the view of B&B as a distinct, platform-level value-creation pathway that generates synergies and consolidation gains beyond organic growth (Hammer et al., 2017; Cohn, Hotchkiss, and Towery, 2022; Bansraj, Smit, and Volosovych, 2020). The unconditional margin advantage of B&B platforms (+4.19 pp) appears to be partly explained by pre-deal selection, as B&B platforms enter PE ownership with different baseline characteristics than standalone investments.

A fundamental limitation of the within-PE B&B analysis is the endogeneity of the B&B treatment variable itself. B&B status is determined by post-investment behaviour: a platform is classified as B&B if it executes at least one confirmed add-on during the observation window. This creates a reverse causality concern that conditioning on pre-deal characteristics only partially addresses. PE-backed companies that are performing well operationally, generating the cash flows and managerial capacity required for acquisition integration, are more likely to pursue add-on acquisitions than underperforming platforms. If this is the case, the estimated B&B revenue premium may partly reflect the superior baseline trajectories of successful PE investments, not the causal effect of the buy-and-build strategy itself. An instrumental variable for B&B activity would be required to establish a clean causal B&B effect, and no such instrument is available in the present data. The B&B sub-analysis should therefore be read as descriptive evidence of two empirically distinct operational profiles within the PE-backed sample, and not as causal estimates.

The most informative evidence emerges from the PSM-based DiD subgroup analysis, which compares each group against its matched control firms, and is presented in Table 12.

Table 12. DiD Subgroup Estimates by B&B Status

Subgroup	N (PE)	ATT Margin (pp)	p	ATT Revenue (pp)	p
B&B platform (≥1 add-on)	77	+0.28	0.888	+32.76***	0.002
Non-B&B (standalone)	149	-4.00***	0.003	+2.62	0.687

*Note: The B&B subgroup DiD estimates are subject to the reverse causality concern discussed in the text: B&B status is determined by post-investment behavior and may partly reflect strong pre-existing performance trajectories and not the effect of add-on activity itself. ATT estimated via PSM-weighted DiD with clustered standard errors. ** p < 0.05, *** < 0.01*

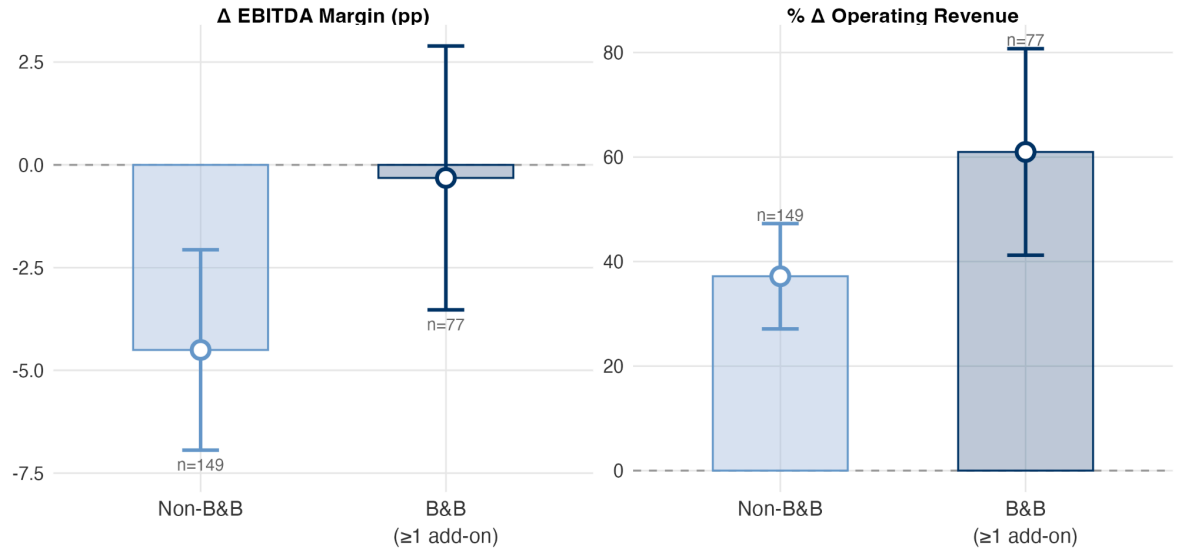
The results point to two different value-creation patterns. Among B&B platforms, PE ownership is associated with strong revenue growth (+32.8 pp above matched controls, p = 0.002), while the margin effect is economically small and statistically insignificant (+0.3 pp, p = 0.888). Among non-B&B platforms, the pattern is reversed. PE ownership coincides with significant margin reduction (-4.0 pp, p = 0.003) but no statistically significant revenue growth (+2.6 pp, p = 0.687). This divergence helps explain the aggregate findings reported in Sections 6.1 and 6.2. The overall negative ATT for EBITDA margin (-2.6 pp) is driven primarily by the non-B&B group, which constitutes 65.9% of the sample, while B&B platforms appear to create value mainly through revenue growth.

These findings are in line with the theoretical distinction between organic operational improvement, which involves restructuring and cost optimisation that may temporarily compress margins, and inorganic growth through consolidation, where revenue scaling via acquisitions dominates the short-term financial profile. The absence of a significant margin effect for B&B platforms may reflect that integration costs and overhead absorption from add-ons offset operational efficiency gains, or alternatively that B&B-focused PE sponsors prioritise top-line growth over short-term margin improvement.

Several limitations should be acknowledged. The B&B classification captures only acquisitions registered under the acquiror's primary BvD ID. Add-ons executed through SPVs or holding companies are not detected, which may understate the true B&B activity. Furthermore, the analysis does not observe the timing of individual add-ons relative to the ±2-year event window, meaning that revenue effects may partly reflect mechanical consolidation of acquired revenues over organic operational

improvement. Future research incorporating deal-level add-on timing would allow sharper identification of the organic versus inorganic components of post-buyout performance.

Figure 10: B&B vs. Non-B&B Operational Performance



Note: Add-on acquisitions identified via Orbis M&A Buy-and-Build filter matched to platform BvD IDs. B&B = platform made ≥1 completed add-on acquisition. Mean change in EBITDA margin (percentage points) and percentage change in operating revenue for PE-backed platform companies with (B&B ≥1 confirmed add-on, n = 77) and without (non-B&B, n = 149) verified add-on acquisitions. 95% confidence intervals. B&B classification based on Orbis M&A Buy-and-Build filter cross-matched to platform company BvD IDs. Post-period: t+1 to t+2, pre-period: t-2 to t-1, t=0 excluded. PE group only.

6.5 Implications for PE Practitioners and Policymakers

The findings of this study have practical implications for both PE fund managers and the policymakers who evaluate the role of PE in the Nordic economies.

For PE practitioners, the evidence of a growth-over-margin pattern suggests that the revenue-first approach often used by Nordic PE sponsors can work, but that it is sensitive to macroeconomic conditions. Deals completed during the high-rate period from 2022 onwards are associated with revenue growth approximately 35 percentage points lower than pre-2022 cohorts, implying that the operational playbooks developed during the low-rate era of 2010 to 2021 require meaningful revision in a structurally higher interest rate environment. PE practitioners should explicitly model the sensitivity of their post-acquisition growth programs to financing cost assumptions and prioritise investments where the revenue growth thesis is less dependent on balance sheet leverage and credit availability.

The OLS results suggest that margin improvement is more likely in larger platform acquisitions, where there is more room for professional management and cost rationalisation initiatives. For smaller targets, the main PE value-add appears to come from revenue growth, not efficiency gains. In these cases, the investment thesis should be assessed over a longer holding period than the ±2-year window used in this study, since short-run costs may appear before the revenue gains are fully realised.

The higher revenue growth linked to institutional buyouts relative to management buyouts highlights the importance of the operational resources and strategic ambition that PE sponsors bring to transactions. This suggests that strong operating partner capabilities and sector-specific expertise remain important sources of advantage in the Nordic mid-market.

For policymakers, the finding that PE ownership is associated with higher revenue growth of +14.3 percentage points relative to matched non-PE-backed Nordic peers in the PSM-DiD framework is consistent with the view that PE ownership may contribute to operational value creation in the Nordic context beyond what firm characteristics alone would predict. However, the precise magnitude should be interpreted alongside the sensitivity analysis in Section 5.4. This result is relevant to the ongoing policy debate about the role of PE in Nordic economies, where PE-backed businesses collectively employ approximately one million people and PE penetration exceeds the level in the rest of Europe (Gain.pro, 2025).

At the same time, the simultaneous margin compression documented in this study serves as a reminder that PE value creation is not synonymous with short-run profitability improvement. Policymakers evaluating PE's societal contribution should consider top-line metrics alongside bottom-line ones and should be cautious about drawing conclusions from either in isolation. The evidence that the post-2021 interest rate environment has constrained the revenue growth capacity of PE-backed companies is also relevant to regulatory discussions about leverage in the PE industry. The results suggest that macroeconomic conditions moderate PE operational outcomes in ways that existing financial stability frameworks may not fully capture.

6.6 Directions for Future Research

Several limitations of this study, discussed in Section 7, have implications for the priorities of future research in this area.

The most significant is the ±2-year event window, which captures only the initial phase of PE ownership. Thomsen and Christoffersen (2024) document that PE-backed firms achieve top-line

growth without corresponding margin or ROA gains in the first half of the holding period, consistent with the possibility that profitability improvements materialise only later in the holding period. Amundsen and Tautra (2025), using a four-year post-treatment window, find that revenue effects accumulate progressively, reaching roughly 25 percent above matched controls by the fourth post-buyout year (T+4). By ending the observation at t+2, this study may also understate the long-run revenue growth linked to PE ownership and may also not fully capture the eventual margin trajectory. Extending the event window to four or five years would represent the most valuable single methodological improvement in future work, and is particularly urgent for the 32 high-rate era deals in the present sample, for which the long-run performance implications of the post-2021 monetary environment remain unclear.

A further limitation is the coverage of the control group. The Orbis-based pool of 4707 Nordic companies is smaller and less representative than the mandatory national accounting registers used in studies such as Amundsen and Tautra (2025). This may reduce the precision of the PSM procedure and leave some residual matching bias that the DiD design cannot fully address. Future research using the Norwegian Corporate Accounts database, Statistics Sweden's firm-level registry, or similar Finnish and Danish sources could address this limitation and produce ATT estimates with greater precision and lower selection bias.

Finally, given the high add-on intensity of the Nordic PE market noted earlier, a meaningful share of the revenue growth documented in this study may reflect inorganic acquisition activity, rather than organic operational improvement. Separating these two sources of growth through deal-level add-on data represents an important direction for future research, as the policy and practitioner implications of organic versus inorganic growth differ considerably. A study that combines the PSM-DiD design employed here with detailed transaction-level data on add-on acquisitions would be well-positioned to separate the two mechanisms and provide a more complete picture of how PE creates value in the Nordic context.

7. Limitations

While this study uses several empirical approaches to address the main identification challenges in the PE performance literature, the results are subject to a number of limitations. These limitations affect how the findings should be interpreted and are discussed below, starting with those that are likely to matter most for the validity of the results.

7.1 Survivorship Bias

The sample construction requires complete financial data across the full ±2-year event window around the buyout date, covering five fiscal years in total (t-2 to t+2). This requirement excludes portfolio companies that may have experienced bankruptcy, forced restructuring, significant ownership changes, or ceased reporting during the observation period. Such firms are likely to be the worst performers within the PE-backed population, and excluding them may bias the sample toward companies with more stable operations and better data availability. This may in turn lead to upward bias in the estimated average treatment effects on both EBITDA margin and revenue growth.

Of the 271 buyouts initially identified in the Orbis database, 45 were excluded due to incomplete data coverage, representing an attrition rate of approximately 17 percent. Whether these excluded observations differ systematically from the retained sample cannot be verified within the scope of this study, and this constitutes one of the most difficult forms of bias to address without access to liquidation or restructuring records. This concern is not unique to the present study. Amundsen and Tautra (2025) acknowledge the same limitation in their Norwegian sample, noting that the mandatory reporting structure of the Norwegian Corporate Accounts database mitigates but does not eliminate survivorship bias.

7.2 Unobserved Confounders and the Limits of Propensity Score Matching

The Propensity score matching (PSM) controls for observable pre-buyout characteristics, specifically firm size (log operating revenue), initial EBITDA margin, and country of origin. This approach reduces selection bias on observable dimensions, as reflected in the post-matching standardised mean differences of -0.120 for log firm size, marginally above the conventional 0.10 threshold recommended by Austin (2011), and -0.030 for initial EBITDA margin. The remaining size imbalance is acknowledged as a minor limitation and is addressed by including log operating revenue directly as a covariate in the DiD specification.

However, PSM cannot address selection on characteristics that are not observed in data. PE fund selection decisions often rely on factors that are not captured in accounting data. Kaplan and Strömberg (2009) describe how PE firms use their industry and operating knowledge to identify attractive investments and develop a value-creation plan at the time of investment, drawing on qualitative factors such as management quality, competitive positioning, and strategic growth potential. These are also factors that may affect long-run operational improvement, but they are not directly visible in financial statement data. Jenkinson et al. (2022) identify this as a central

methodological challenge in the field: if PE funds select companies that would have outperformed regardless of ownership, the estimated Average Treatment Effect on the Treated (ATT) will be upwardly biased even after matching.

This concern applies with different force to the two estimates. As the Rosenbaum sensitivity analysis in Section 5.4 shows, the margin ATT (-2.6 pp) is robust to substantial unobserved confounding (significant through Γ = 1.5), whereas the revenue ATT (+14.3 pp) is considerably more fragile, losing significance at Γ = 1.1. The revenue ATT should therefore be read as an upper bound on the causal PE effect, while the margin ATT can be interpreted with greater confidence. The DiD design accounts for time-invariant unobserved differences through the within-firm differencing structure, but time-varying unobservables, such as strategic initiatives initiated prior to but materialising after the buyout, cannot be eliminated.

The low explanatory power of the cross-sectional OLS models (R^2 = 0.067-0.081) also points to the importance of unobserved differences in post-buyout outcomes. While the coefficients on the primary variables of interest are statistically significant and stable across specifications, the models leave the substantial majority of outcome variance unexplained. This limits the study's ability to characterise the full set of factors that shape post-buyout performance. It also suggests that variables not available in Orbis, such as PE fund characteristics, management team quality, sector-specific competitive dynamics, and deal structuring details, matter for the observed outcomes. Future research using fund-level data or survey-based operational metrics could improve explanatory power beyond what accounting-based cross-sectional designs can achieve.

7.3 Short Event Window and the Truncation of Long-Run Effects

The symmetric ±2-year event window in this study captures operational performance in the two fiscal years prior and following the buyout date, with the transaction year excluded. While this window is useful for documenting short-run post-acquisition changes, it may not capture the full longer-run development of operational performance. The PE value creation literature suggests that structural operational improvements, especially those involving strategic repositioning, buy-and-build execution, and management professionalisation, often require three to five years to materialise fully in accounting metrics. Amundsen and Tautra (2025) employ a four-year post-treatment window (T-3 to T+4) and document that revenue effects build over time, with treated firms reaching roughly 25 percent above matched controls by year four (T+4).

By shortening the observation window at t+2, this study may understate the long-run revenue growth connected to PE ownership. It may also mischaracterise the margin trajectory: short-run margin decline driven by investment-phase costs could reverse into margin expansion over a longer horizon that remains outside the observation window. This concern is particularly relevant for the 32 high-rate era deals in the sample, which were completed from 2022 onwards and for which t+2 falls at or near the end of the data coverage period.

7.4 SPV/Bidco Debt Structures and the Measurement of Pre-Buyout Leverage

Leveraged buyouts are typically structured using a special purpose vehicle (SPV) or acquisition holding company (Bidco/Newco) that raises the acquisition debt. The operating portfolio company, which is the entity tracked in the Orbis database and measured in this study, frequently appears with low reported debt on its balance sheet, even when the consolidated group is highly leveraged. This structure causes the debt-to-assets ratio at t0, constructed from Orbis balance sheet data using shareholders' funds and total assets, to systematically understate the true LBO leverage embedded in the capital structure at the time of acquisition.

The degree of understatement varies across deals. Some PE funds push acquisitions debt down to the operating company level (“debt pushdown”), while others retain it entirely at the Bidco level. As a result, two deals with identical economic leverage may appear to have notably different leverage ratios in Orbis. Axelson et al. (2013) document that buyout leverage is driven more by credit market conditions than by company fundamentals, implying that this measurement error is not random but potentially correlated with deal timing. Deals completed in favorable credit environments may carry higher true leverage than the Orbis data suggests. Jensen's (1989) “debt as discipline” channel, where leverage constrains managerial discretion and drives operational improvement, may therefore be stronger than what is captured by the leverage control included in the extended OLS specifications.

7.5 Control Group Coverage and Orbis Representativeness

The external control group of 4707 Nordic companies collected from Orbis represents a much smaller and potentially less representative pool than the full universe of Nordic private companies. This stands in contrast to studies exploiting mandatory national accounting registers, such as Amundsen and Tautra (2025), who use the Norwegian Corporate Accounts database with more than 831 000 company-year observations, and near complete coverage of the Norwegian private sector.

Orbis coverage of Nordic private companies is subject to known selection biases. Larger, fast-growing, and more financially sophisticated firms are more likely to appear in the database, and coverage rates

vary systematically across countries over time (Kalemli-Özcan et al., 2024). Although holding company and financial intermediary entities were excluded from the control pool using NACE code filters (code 6420, 6810-6820, and related financial sector codes), the remaining pool may still overrepresent certain firm types relative to the broader population of PE-comparable Nordic companies.

The practical implication is that the PSM matching procedure selects from a restricted and potentially non-representative candidate set. As a result, matched controls may still differ from PE targets on dimensions that are not fully captured by observable balance tests. The robustness checks in Section 5.2, which repeat the main PSM-DiD estimates using a stricter caliper of 0.15 standard deviations, help assess the sensitivity of the ATT estimates to matching specifications, but cannot address the representativeness of the underlying control pool.

7.6 The Parallel Trends Assumption

The validity of the DiD estimator depends on the parallel trends assumption. In the absence of PE ownership, treated and control firms would have followed similar performance paths over the ±2-year observation window. Because the counterfactual outcome is not observable, the assumption can only be assessed indirectly through pre-treatment patterns (Angrist & Pischke, 2009). The study therefore assesses parallel trends using two pre-treatment checks. First, a t-test comparing pre-buyout EBITDA margin trends between matched treated and control firms gives a p-value of 0.843, suggesting no statistically significant difference in pre-period trends. Second, a placebo regression assigning a fictitious treatment date within the pre-period produces an insignificant treatment coefficient ($p = 0.212$), providing further support for the assumption. These checks support the parallel trends assumption, but they do not prove it. A remaining concern is that PE funds may acquire companies just before an organic improvement, for example, after a new management hire or product launch. In that case, post-buyout acceleration could reflect pre-deal developments, as opposed to PE ownership itself (Boucly et al., 2011). This type of time-varying selection bias cannot be addressed through PSM or DiD without an instrument for PE investment timing.

7.7 The Stable Unit Treatment Value Assumption

The DiD framework assumes that the treatment of one firm does not affect the outcome of control firms, known as the stable unit treatment value assumption (SUTVA). In concentrated Nordic industry segments, this assumption may be violated. PE-backed portfolio companies that pursue buy-and-build strategies, enter new markets, or compete on price after a well-capitalised acquisition may directly affect the revenue growth and margins of non-PE-backed competitors that appear in the

control group. If control firms are put at a disadvantage by the competitive activities of PE-backed firms in the same sector, the ATT estimate for revenue growth would be upwardly biased. In that case, part of the estimated treatment effect could reflect competitive pressure on control firms instead of operational value creation within the PE-backed firms themselves. This concern is relevant in the Nordic PE market, where buy-and-build activity is widespread (Gain.pro, 2025). While sector fixed effects in the DiD specification control for average differences across sectors, they cannot capture firm-level spillover effects from PE-backed competitors.

7.8 Geographic Coverage and the Underrepresentation of Denmark

Although this thesis studies the Nordic PE market, the Danish sub-sample comprises only 16 observations, representing 7 percent of the total. This stands in contrast to Sweden (39 percent), Finland (33 percent), and Norway (20 percent), and reflects both the historically smaller scale of the Danish institutional buyout market and the more limited Orbis coverage of Danish private companies relative to mandatory national registers such as the Danish CVR system. The country fixed effects included in both the OLS and PSM-DiD specifications control for average performance differences linked to Denmark, but the small sub-sample means that Danish-specific dynamics, including the role of Danish PE sponsors, sectoral composition, and the Danish regulatory environment, are unlikely to be meaningfully identified in the cross-sectional results. The findings of this study should therefore be interpreted with particular caution when applied to the Danish context, and future research drawing on mandatory Danish accounting registers would be better positioned to provide Denmark-specific evidence. This limitation also implies that the "Nordic" framing of the study is mainly driven by the Swedish, Finnish, and Norwegian PE markets, which should be kept in mind when interpreting the aggregate findings.

Table 13: Summary of Study Limitations

Limitation	Description	Severity
Survivorship bias	Selection on data completeness (±2 years) likely excludes distressed firms and bankruptcies, skewing the sample toward stable performers.	**Moderate - High**
Unobserved confounders	PSM controls for observables only: managerial quality, strategic positioning, and PE fund experience remain unobserved.	**High**

Short event window (±2 years)	Post-buyout operational changes may materialize over longer horizons. Long-run effects (t+3,t+4) are not captured.	**Moderate**
SPV/Bidco debt structure	LBO debt is typically held at the holding company level (Bidco/Newco), causing Orbis to systematically understate portfolio company leverage at t0.	**Moderate**
Control group coverage	Orbis control pool (4707 companies) is substantially smaller and less representative than full national business registers, potentially introducing matching bias.	**Moderate**
Parallel trends	DiD relies on the parallel trends assumption, which cannot be verified directly. Pre-trend diagnostics (p=0.843; placebo p=0.212) support it.	**Low-Moderate**
SUTVA	Treatment of one firm may affect control firms (e.g., PE-backed competitors gain market share), violating the stable unit treatment value assumption.	**Low**
Denmark underrepresentation	Denmark accounts for only 16 observations (7%), limiting the generalisability of findings to the Danish PE market despite its inclusion in the "Nordic" sample frame.	**Low-Moderate**

Overall, these limitations mean that the findings of this study should be read mainly as evidence of an association between PE ownership and operational performance in the Nordic context, not as precise causal estimates. The combination of PSM-based covariate balancing and the DiD differencing structure represents the most credible identification strategy available within the constraints of the Orbis data environment, but it cannot provide the same clean causal identification as an instrumental-variable or regression-discontinuity design. Future research using mandatory national accounting registers, which offer near-complete population coverage and higher data quality than commercial databases, could better address the survivorship bias and control group representativeness limitations identified above. Extending the event window to four or five

post-buyout years, and modeling buy-and-build activity using deal-level add-on data would also improve the design used in this study.

8. Conclusion

This thesis examined whether private equity buyouts are associated with measurable operational changes in Nordic portfolio companies, and which deal-level and macroeconomic factors explain differences in post-buyout performance. Using a sample of 226 completed buyouts across Sweden, Norway, Denmark, and Finland from 1999 to 2023, the study applied a three-stage empirical strategy combining before-after univariate analysis, OLS cross-sectional regressions with HC1 robust standard errors, and a propensity score matching and difference-in-differences framework against matched controls drawn from a pool of 4707 comparable non-PE-backed Nordic companies. The sample is drawn from all four Nordic countries, but the limited Danish coverage means the findings are best read as evidence on the Swedish, Finnish and Norwegian PE markets.

The analysis points to five main findings that together give a coherent picture of PE operational strategy in the Nordic context. First, PE buyouts are linked to a statistically significant mean decline in EBITDA margins of 3.08 percentage points alongside significant revenue growth, a mean increase of approximately 45 percent and a median increase of approximately 30 percent in the post-buyout period. This pattern confirms the theoretical prediction advanced in H1. Following Boucly, Sraer, and Thesmar (2011), the hypothesis anticipated that PE ownership would relax financial constraints and enable investment programs that reduce short-run margins while supporting top-line trajectories. The Nordic PE market appears to be characterised by a growth-over-margin pattern rather than the efficiency extraction that dominated the early U.S. buyout literature.

Second, buyouts completed in 2022 or later are related to approximately 35 percentage points lower post-buyout revenue growth relative to pre-2022 cohorts, making this the largest effect estimated in the study. This finding suggests that the post-pandemic monetary tightening cycle has constrained the revenue growth capacity of PE-backed portfolio companies in the Nordic context. The coefficient is significant at the one percent level in the baseline and leverage-augmented OLS specifications, but its effect is sensitive to the choice of comparison period and it is not robust to separating the 2020-2021 COVID-era cohort from the reference group. This finding therefore has potential implications for PE practitioners whose growth strategies were developed during the low-rate period from 2010 to 2021, though it should be interpreted with caution. The subgroup DiD analysis directionally confirms this result, with the high-rate cohort exhibiting a revenue ATT approximately -18 percentage points, compared to +17.66 percentage points for the pre-2022 cohort, though this

estimate does not reach conventional significance levels (p = 0.257), reflecting the limited statistical power available in the high-rate subsample of 32 matched observations. H2 should therefore be viewed as directionally supported. The high-rate cohort consistently shows a large negative revenue effect, but its precise scale and significance are sensitive to the comparison period and the small high-rate subsample, and the causal interpretation remains directional, and not conclusive.

Third, institutional buyouts are associated with approximately 39 percentage points higher post-buyout revenue growth than management buyouts. This is supportive of the prediction that PE sponsors in IBOs bring stronger operational resources, greater strategic ambition, and a more active growth approach than what is typically available in management-led transactions.

Fourth, larger firms at the time of acquisition experience significantly greater EBITDA margin improvement post-buyout, at approximately 1.69 percentage points per log unit of pre-buyout revenue, consistent with PE operational engineering producing scale-conditional returns in the Nordic private-to-private context. Smaller targets appear equally capable of generating revenue growth under PE ownership, but the margin improvement connected with professional management frameworks and cost rationalization appears to be concentrated in larger firms with more developed organisational structures.

Fifth, the PSM-DiD analysis shows that PE-backed Nordic companies experience EBITDA margin compression of approximately 2.6 percentage points relative to their propensity-score-matched non-PE-backed controls, while achieving approximately 14.3 percentage points higher revenue growth. Both estimates are statistically significant at the five percent level. However, these two estimates differ in how robust they are to hidden bias. The margin ATT is robust to unobserved confounding through $\Gamma = 1.5$, indicating that an unobserved confounder would need to increase the odds of PE selection by about 50 percent before the margin result loses statistical significance. The revenue ATT, by contrast, loses significance at $\Gamma = 1.1$, meaning that a relatively modest degree of hidden bias could explain away the positive revenue effect. The revenue ATT should therefore be interpreted as an upper bound on the causal PE revenue effect, not as a precise estimate, while the margin ATT is the more robust of the two PSM-DiD findings.

This thesis makes three contributions to the literature. It provides empirical evidence on Nordic PE operational performance using a sample that extends to 2023, thereby broadening the time coverage of existing Nordic-specific studies and capturing the full arc from the low-rate era through the post-pandemic monetary tightening cycle. It is, to the authors' knowledge, the first study to explicitly test the role of the post-2021 interest rate environment as a moderator of PE operational

performance in the Nordic context, adding a macroeconomic dimension to a literature that has largely assumed accommodative financing conditions. And it employs a causal identification strategy, propensity score matching combined with difference-in-differences estimation, that addresses the selection bias concern that is central to the PE performance literature and that purely descriptive before-and-after comparisons cannot resolve.

Several limitations affect how these findings should be interpreted. The ±2-year event window captures only the initial phase of PE ownership and may understate the long-run revenue growth attributable to PE sponsorship. It may also fail to capture the full margin trajectory if investment-phase costs reverse in later years. Survivorship bias from the requirement of complete financial data across the five-year observation window means that the ATT estimates reported here should be interpreted as upper bounds on the causal PE effect for the full population of Nordic buyouts. The Orbis-based control group, while adequate for the purposes of this study, is less representative than the mandatory national accounting registers available in some Nordic countries. Finally, the inability to separate organic from inorganic revenue growth limits the conclusions that can be drawn about the specific mechanisms through which PE creates revenue growth.

These limitations also suggest clear directions for future research. Extending the event window to four or five post-buyout years, using mandatory national accounting registers for more representative control groups, and integrating deal-level add-on transaction data to separate organic and inorganic growth would improve on the design used in this study. The post-2021 high-rate cohort of 32 deals in the present sample deserves particular attention in future work. As these companies move through their full holding periods and exit cycles, it will become clearer whether the revenue growth constraint documented here is temporary or more persistent. This will matter for both PE practitioners operating in a higher-rate environment and policymakers assessing the long-run contribution of PE to the Nordic economies.

For PE practitioners, the findings suggest that growth strategies developed during the low-rate era of 2010 to 2021 may need to be reassessed in a higher interest rate environment. Strategies that rely on balance sheet leverage and cheap credit to fund market expansion, headcount investment, and buy-and-build activity face structurally less favorable conditions when financing costs rise sharply. Practitioners should therefore stress-test post-acquisition growth plans against higher-for-longer rate scenarios and prioritise investment theses where revenue growth is less dependent on credit availability.

For policymakers, the estimated revenue outperformance of +14.3 percentage points relative to matched non-PE-backed peers provides evidence that PE ownership may contribute to measurable revenue growth in the Nordic economies. At the same time, the value creation appears sensitive to the macroeconomic environment. This suggests that assessments of PE's role in the Nordic economies should account not only for average performance effects, but also for how those effects vary across financing conditions.

Appendix

Table A.1: OLS Regression Results - %ΔEBITDA

	(1) Baseline	(2) + Leverage
High-rate era (≥2022)	-46.420** (22.022)	-48.739** (22.217)
IBO Dummy	5.964 (51.180)	13.216 (52.476)
Log(Op. Revenue)t0	7.913 (6.727)	6.212 (6.938)
Initial EBITDA Margin	0.569 (0.422)	0.684 (0.426)
Low-rate era (2017-2019)	-8.847 (21.974)	-7.435 (21.728)
Debt ratio t0	-	77.880 (49.661)
Country FE / Sector FE	Yes / Yes	Yes / Yes
Observations	226	226
R^2 / Adj. R^2	0.062 / -0.010	0.077 / 0.001

*Note: Dependent variable: percentage change in EBITDA (post minus pre). HC1 robust standard errors in parentheses. Reference categories: Sweden (Country), Other services (Sector). Debt ratio winsorized at p1/p99. *p<0.10, **p<0.05,***<0.01.*

Table A.2: Variable Definitions

Variable	Definition	Unit	Source
Δ EBITDA Margin	Avg. post-buyout margin (t+1,t+2) - avg. pre-buyout margin (t-2,t-1)	pp	Orbis
%Δ Operating Revenue	% change in operating revenue, post- vs. pre-buyout average	%	Orbis
%Δ EBITDA	% change in EBITDA, post- vs. pre-buyout average	%	Orbis

Log(Op. Revenue)t0	Natural log of operating revenue in the deal year	log EUR	Orbis
Initial EBITDA Margin (t0)	EBITDA margin in the deal year	%	Orbis
Debt Ratio (t0)	Total debt / total assets at t0, winsorised at p1/p99	ratio	Orbis
IBO dummy	1 = institutional buyout, 0 = management buyout	{0,1}	Orbis M&A
High-rate era	1 = deal completed in 2022 or later	{0,1}	Orbis M&A
Low-rate era	1 = deal completed 2017-2019	{0,1}	Orbis M&A
Treated	1 = PE-backed firm, 0 = matched control	{0,1}	Orbis M&A
%Δ Revenue per employee	% change in (operating revenue/employees), post vs. pre	%	Orbis
B&B dummy	1 = platform with ≥1 add-on acquisition	{0,1}	Orbis M&A
Add-on count	Number of add-on acquisitions by the platform	count	Orbis M&A

Note: All outcomes use a symmetric ±2-year event window (pre = average of t-2, t-1, post = average of t+1,t+2). Country, year, and sector fixed effects are included.

Table A.3: Propensity Score Model (Logit)

	Dependent Variable:
	PE Treatment (1 = PE-backed)
log_size	-0.260***
	(0.044)
margin_t0	0.007*
	(0.004)
country_feNO	-1.636***
	(0.187)
country_feDK	0.071
	(0.277)
country_feFI	-0.969***
	(0.164)
Constant	-0.913*
	(0.478)

Observations	17 685
Log Likelihood	-1157.146
Akaike Inf. Crit.	2326.292

*Note: Logit model used to estimate propensity scores, regressing PE treatment status on log operating revenue, initial EBITDA margin, and country fixed effects. As described in Section 3.7, matching is performed within deal-year cohorts, the 17 685 observations include the 226 treated firms and 17 459 control observations drawn from the 4707-firm cleaned control pool. Matching produces the 904-observation matched sample used in the DiD analysis. Reference country: Sweden. *p<0.1, **p<0.05, ***p<0.01*

Table A.4: Labour Productivity Regressions

	Dependent Variable:	
	% Change Employees	% Change Revenue per Employee
	(1)	(2)
margin_t0	0.152	-0.037
	(0.216)	(0.172)
log_size	-2.871	-3.224
	(3.332)	(2.329)
IBO	33.580*	-12.544
	(19.303)	(26.677)
high_rate	-39.772***	-2.044
	(11.497)	(8.385)
low_rate_era	-6.204	1.283
	(11.373)	(6.616)
Constant	-30.444	139.982***
	(32.517)	(36.437)
Country FE	Yes	Yes
Sector FE	Yes	Yes
Observations	219	219
R^2	0.085	0.083
Adjusted R^2	0.013	0.011
Residual Std. Error (df = 202)	66.338	46.955

F Statistic (df = 16; 202)	1.178	1.147

*Note: HC1 robust standard errors. Sample 219 buyouts with employee data. *p<0.1,**p<0.05,***p<0.01*

Table A.5: Pairwise Correlations Among Covariates

	log_size	margin_t0	debt_ratio_t0
log_size	1	0.233	0.083
margin_t0	0.233	1	-0.109
debt_ratio_t0	0.083	-0.109	1

Note: Pearson pairwise correlation coefficient among the covariates used in the PSM-DiD specifications. All pairwise correlations are below 0.25 in absolute value, indicating that multicollinearity is not a concern in the regression models.

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AI Declaration

In this thesis, we have used AI-based tools, specifically Claude (Anthropic) and ChatGPT (OpenAI), as support tools for language editing, proofreading, and improving clarity. We also used these tools to receive suggestions for R code and to help identify and debug coding errors. All empirical analysis, methodological choices, coding decisions, interpretation of results, and final written content are our own.